DETECTION OF RECRUITMENT FRAUD USING HYBRID TEXT INTELLIGENCE AND TRUST SCORING

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

In the rapidly expanding digital recruitment ecosystem, the rise of fraudulent job postings has become a significant concern, causing financial loss, identity theft, and exploitation of job seekers. Traditional methods of detecting fake jobs often rely on manual review or user reporting, which can be time-consuming, inconsistent, and prone to error. To address this challenge, this project proposes an intelligent Recruitment Fraud Detection System using machine learning and a Hybrid Decision Engine for automated verification. The system classifies job postings into Legitimate, Fake, or Suspicious categories by analyzing textual content and metadata attributes. It follows a structured workflow comprising Data Preprocessing, Machine Learning Classification, Hybrid Decision Analysis, and Company Verification. In preprocessing, job descriptions and associated features are cleaned, tokenized, and vectorized using TF-IDF techniques. The processed data is then passed through supervised machine learning models such as Logistic Regression and Random Forest, trained to detect patterns indicative of fraudulent postings. The Hybrid Decision Engine consolidates model predictions with rule-based validations to enhance accuracy and reduce false positives. Additionally, the system performs company verification by analysing available organizational information and related job references to assess credibility. Implemented in Python, Scikit-learn, and Streamlit, the system provides a real-time interface for job analysis and automated report generation. Overall, the proposed system helps users identify fraudulent job postings efficiently, promotes trust in online recruitment platforms, and supports safer, more informed career decisions.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
LR	Logistic Regression
NLP	Natural Language Processing
TF-IDF	Term Frequency – Inverse Document Frequency
CSV	Comma Separated Values
GPU	Graphics Processing Unit

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In the modern digital recruitment landscape, online job portals and professional networking platforms have transformed the way employers and job seekers connect. While this digital transformation has increased accessibility and employment opportunities, it has also led to a significant rise in fraudulent job postings. Cybercriminals exploit online recruitment systems by publishing deceptive advertisements to collect personal information, demand registration fees, or conduct financial scams. As a result, job seekers face serious risks including identity theft, monetary loss, and data misuse. Traditional manual verification mechanisms are insufficient to effectively monitor and filter the vast volume of job postings generated daily.

This project addresses these challenges by introducing a Machine Learning-based Recruitment Fraud Detection System integrated with a Hybrid Decision Engine to ensure intelligent, automated, and reliable identification of fraudulent job advertisements. The proposed system leverages Natural Language Processing (NLP) techniques and supervised machine learning algorithms to analyse textual job descriptions, company details, and metadata attributes. By learning patterns from historical datasets containing both legitimate and fake postings, the system can accurately classify new job posts into Legitimate, Fake, or Suspicious categories.

The architecture of the proposed system consists of multiple core modules, including Data Preprocessing, Feature Extraction, Machine Learning Classification, Hybrid Decision Analysis, and Company Verification. During

preprocessing, raw job text is cleaned, tokenized, and transformed using TF-IDF vectorization to convert textual information into numerical feature representations. These features are then processed by machine learning models such as Logistic Regression and Random Forest to detect fraudulent characteristics. The Hybrid Decision Engine enhances system robustness by combining model predictions with additional rule-based validations, thereby reducing misclassification and improving reliability.

Furthermore, the system incorporates a company verification component to analyse the credibility of organizations associated with job postings. This additional layer of validation increases trust and minimizes false legitimacy claims. The complete application is implemented using Python, Scikit-learn, and Streamlit, enabling real-time analysis, user interaction, and automated reporting.

By integrating machine learning algorithms with hybrid validation mechanisms, this project provides a scalable and efficient framework to combat online recruitment fraud. It enhances transparency, protects job seekers from cyber threats, and contributes toward building a safer digital employment ecosystem.

1.2 PROBLEM DEFINITION

In today's digital recruitment ecosystem, the widespread use of online job portals and professional networking platforms has significantly increased employment opportunities. However, this rapid expansion has also led to a substantial rise in fraudulent job postings. Cybercriminals exploit digital platforms to publish deceptive job advertisements that mislead job seekers into providing sensitive personal information, paying fraudulent registration fees, or disclosing financial details. These scams result in identity theft, monetary loss,

and psychological distress, creating serious concerns regarding trust and safety in online recruitment systems.

Traditional verification mechanisms used by job portals are often insufficient to detect sophisticated fraudulent patterns, especially when fake postings closely resemble legitimate advertisements. Manual moderation is time-consuming, inconsistent, and impractical due to the massive volume of job posts generated daily. Moreover, existing systems primarily rely on basic keyword filtering or user reports, which fail to provide proactive and intelligent detection. The absence of automated, data-driven fraud detection frameworks increases the risk of fake job advertisements remaining active for extended periods.

Additionally, distinguishing between legitimate, suspicious, and clearly fraudulent job postings presents a significant technical challenge. Fraudulent posts may mimic real company names, use professional language, and appear authentic, making detection complex without advanced analytical techniques. There is also a lack of integrated systems that combine machine learning-based classification with supplementary validation mechanisms such as company credibility verification.

Therefore, there is a pressing need for an intelligent, automated, and scalable solution capable of accurately identifying fraudulent job postings in real time. The system must analyse textual content and related attributes using advanced machine learning techniques while incorporating hybrid validation strategies to enhance reliability. Addressing these challenges is essential to protect job seekers, improve trust in online recruitment platforms, and reduce employment-related cyber fraud in the digital environment.

CHAPTER 2

LITERATURE SURVEY

This literature survey gives an overview of important research contributions in Recruitment Fraud Detection System. It focuses on privacy-preserving AI, local model inference, and hybrid retrieval strategies. Each cited paper offers insights into the developing technologies that support secure, air-gapped document intelligence without depending on external APIs.

1. Naudé, M., Adebayo, K. J., and Nanda, R. (2023). "A machine learning approach to detecting fraudulent job types. "In this study, the authors explore the growing misuse of online recruitment platforms through fraudulent job postings. They implement supervised machine learning classifiers such as Logistic Regression, Support Vector Machine, and Random Forest to automatically identify fake job types. The paper uses job description text and company profile features as input and evaluates model performance using accuracy, precision, recall, and F1-score. Key Contribution: Demonstrates a comparative analysis of multiple classifiers to identify fraudulent job postings effectively. Limitation: The approach relies on manually engineered features and lacks deep semantic understanding of complex scam patterns.

2. Saha, S., Mukherjee, A., and Singh, P. K. (2020). "Enhancing the detection of deceptive job advertisements using machine learning techniques. This research addresses the problem of online job fraud by developing machine learning models capable of distinguishing genuine from deceptive job ads. Text preprocessing and TF-IDF vectorization are applied before training classifiers including Random Forest and Support Vector Machine. The system is evaluated using standard metrics such as accuracy and precision. Key Contribution: Improves the detection of deceptive job postings using enhanced feature extraction methods. Limitation: The method may not

adapt well to dynamically changing fraudulent patterns.

3. Kumar, R., Verma, S., and Sharma, A. (2021). "Identification of fake job recruitment using machine learning models." The authors investigate automated detection of fraudulent job postings on online platforms. They implement Naïve Bayes, Logistic Regression, Support Vector Machine, and Random Forest classifiers using job description text processed into numerical features. Evaluation is performed using accuracy and F1-score. Key Contribution: Provides a systematic comparison of multiple supervised learning models for fake job recruitment detection. Limitation: The study does not explore deep learning or explainable AI techniques for enhanced interpretability.

4. Patel, R., Shah, A., and Mehta, N. (2020). "Detection of fake job posts using machine learning algorithms." This paper tackles the growing threat of online recruitment fraud using supervised machine learning techniques. Preprocessing of job descriptions and TF-IDF vectorization is followed by classification with Naïve Bayes, Support Vector Machine, and Random Forest. System performance is measured with accuracy, precision, and recall, highlighting Random Forest as the most effective classifier. Key Contribution: Offers a simple and deployable model for fake job post detection. Limitation: Lacks contextual understanding and struggles with evolving fraud strategies.

5. Reddy, K., Kumar, P., and Rao, S. (2021). "Fake job detection in online recruitment platforms using machine learning." The study presents a framework for early identification of fraudulent job postings by combining structured company information with unstructured job description text. Logistic Regression and Support Vector Machine classifiers are applied after feature extraction and preprocessing. Key Contribution: Demonstrates an efficient machine learning framework applicable to real-world recruitment platforms.

Limitation: Relies only on traditional machine learning methods without leveraging deep learning or explainability.

6. Sharma, A., Malhotra, R., and Gupta, V. (2022). "Fake job post detection using deep learning techniques." This research addresses limitations of traditional methods by employing deep neural networks for fake job detection. Job description text is converted into word embeddings and fed into a deep learning model. Evaluation uses accuracy, precision, recall, and F1-score metrics. Key Contribution: Captures complex semantic relationships in job postings for improved detection accuracy. Limitation: Requires large labeled datasets and high computational resources, with limited interpretability.

7. Zhang, Y., Wang, L., and Chen, H. (2020). "An ensemble learning approach for fraudulent job advertisement detection." The paper presents an ensemble of machine learning models including Random Forest, Gradient Boosting, and Support Vector Machine to improve detection robustness. Both job description text and metadata attributes are used as input features. System performance is evaluated using accuracy, precision, recall, and F1-score. Key Contribution: Reduces overfitting and enhances stability compared to individual models. Limitation: Ensemble methods increase computational complexity and reduce interpretability.

8. Alghamdi, A., Alharbi, F., and Khan, S. (2019). "NLP-based detection of scam job advertisements. This study focuses on using natural language processing techniques to identify fraudulent job postings. Text preprocessing, n-gram feature extraction, and TF-IDF vectorization are combined with linear classifiers such as Logistic Regression. Accuracy and F1-score are used for evaluation. Key Contribution: Demonstrates a lightweight, text-only approach suitable for computationally constrained environments. Limitation: Does not capture contextual semantics and may underperform on complex or well-crafted scam ads.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the current online recruitment ecosystem, most job portals rely on manual moderation, basic keyword filtering, and user-reported complaints to identify fraudulent job postings. These traditional approaches are largely reactive and lack intelligent automation, allowing many fake job advertisements to remain active until users report them. Existing systems typically do not employ advanced Machine Learning or Natural Language Processing techniques to analyze job descriptions in depth, nor do they provide strong company credibility verification mechanisms. As a result, detection accuracy is limited, false classifications are common, and job seekers remain vulnerable to financial scams, identity theft, and misleading employment opportunities.

Techniques:

- Manual Verification
- Basic Keyword Filtering
- User Complaint-Based Reporting
- Simple Rule-Based Checks

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Demerits:

1. Lack of automated intelligent fraud detection
2. High dependency on manual moderation
3. Delayed identification of fake job posts
4. High probability of false positives and false negatives
5. Limited company credibility verification

3.2 PROPOSED SYSTEM

The proposed system introduces a Machine Learning-based Fake Job Post Detection framework integrated with a Hybrid Decision Engine to provide accurate and automated fraud detection. The system analyzes job descriptions, company information, and related metadata using Natural Language Processing techniques such as text preprocessing and TF-IDF vectorization. Supervised models like Logistic Regression and Random Forest are trained to identify fraudulent patterns, and their outputs are combined to enhance prediction reliability. Additionally, a company verification module strengthens credibility assessment, while a Streamlit-based interface enables real-time job analysis and report generation. This approach ensures improved accuracy, reduced fraud risk, and enhanced trust in digital recruitment platforms.

3.2.1 Techniques:

- Machine Learning Algorithms (Logistic Regression, Random Forest).
- Natural Language Processing (Text Cleaning, Tokenization, TF-IDF).
- Hybrid Decision Engine.
- Company Verification Module.
- Python, Scikit-learn, Streamlit.

3.2.2 Advantages of the Proposed System:

1. Automated and real-time fraud detection.
2. Higher accuracy through hybrid model integration.
3. Classification into Legitimate, Fake, and Suspicious categories.
4. Reduced financial and identity fraud risks.
5. Scalable and integrable with online recruitment platforms.

3.3 DEVLOPMENT ENVIRONMENT

3.3.1 SOFTWARE REQUIREMENTS

- FRAMEWORK : Streamlit
- LANGUAGE : Python
- LIBRARIES : Scikit-learn, Pandas, NumPy
- DATA VISUALIZATION : Matplotlib
- MODEL STORAGE : Joblib / Pickle
- PDF GENERATION : ReportLab
- OPERATING SYSTEM : WINDOWS 10
- IDE : Jupyter Notebook
- WEB BROWSER : Google Chrome

3.3.2 HARDWARE REQUIREMENTS

- PROCESSOR : Intel Core i3 or Higher
- RAM : 4GB DDR RAM or Higher
- HARD DISK : 500GB / 1TB
- SYSTEM TYPE : 64-bit System

CHAPTER 4

SYSTEM DESIGN

4.1 UML DIAGRAMS

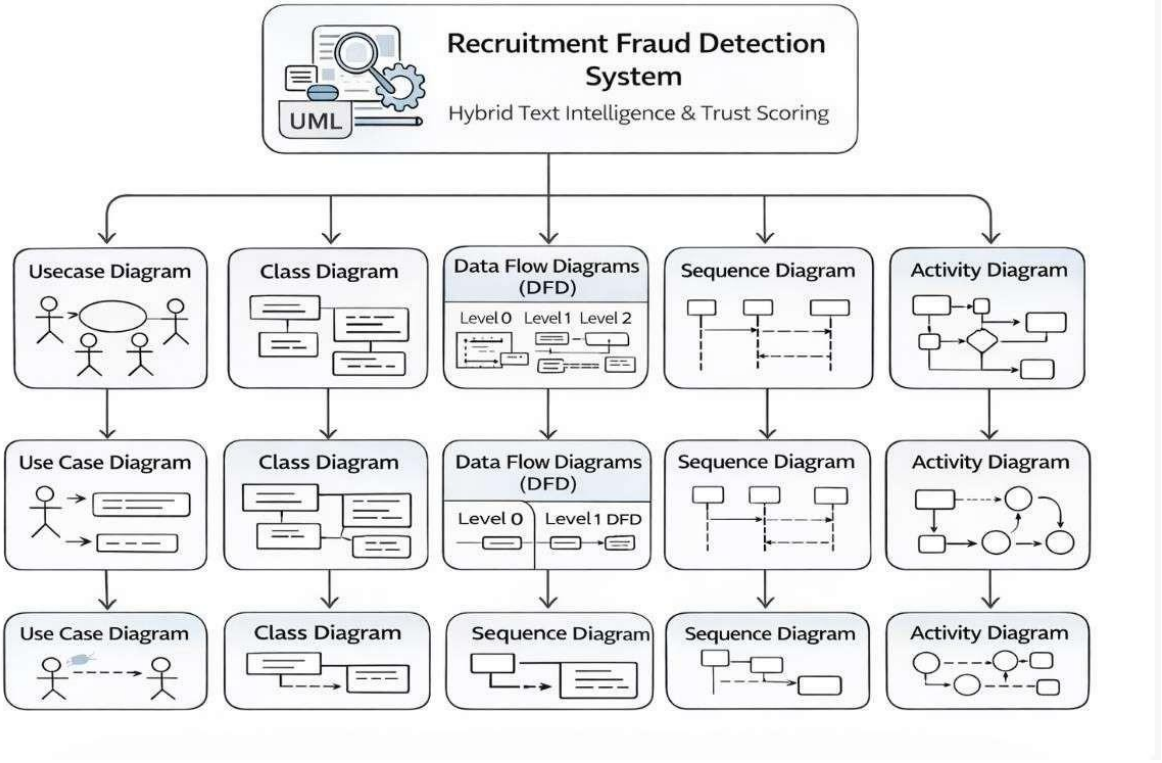


Fig 4.1 UML diagram for Recruitment Fraud Detection System

The UML diagrams provide a structural and behavioural representation of the Recruitment Fraud Detection System using Hybrid Text Intelligence and Trust Scoring. The system mainly consists of modules such as user management, job post processing, text analysis, company verification, and fraud detection. The Use Case Diagram illustrates how users interact with the system to submit job postings and view fraud analysis results. The Class and Package diagrams describe the internal structure of system components including the job

post module, text processing module, fraud detection model, and reporting system. The workflow of the system from job submission to fraud detection, trust score calculation, and report generation is represented through Sequence, Activity, and Collaboration diagrams. These UML representations help visualize how different modules interact to analyse recruitment data and identify fraudulent job postings.

4.1.1 USECASE DIAGRAM

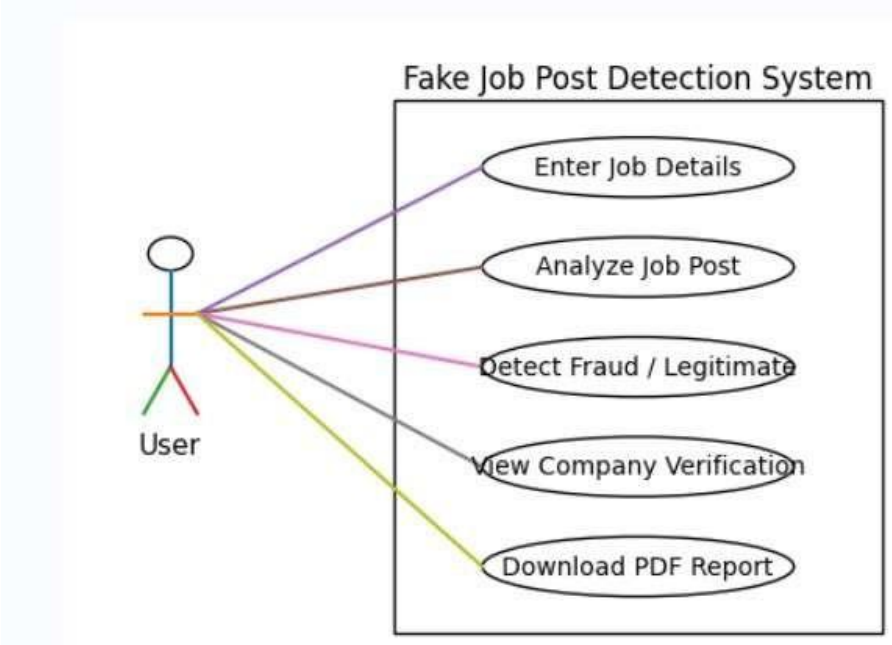


Fig 4.1.1 Use case diagram for Recruitment Fraud Detection System

The Use Case Diagram for this project illustrates the interaction between the User and the Recruitment Fraud Detection System, which analyzes job postings to determine whether they are legitimate, suspicious, or fraudulent. The process begins when the user accesses the system through the Streamlit web interface and enters job details such as job title, company name, job description, and location. The system then performs text preprocessing and feature extraction to prepare the input data for analysis. These features are passed to the trained machine learning model, which identifies patterns that may indicate

fraudulent recruitment activity. In addition, the system performs company verification by retrieving related company information and job links to support the analysis. Based on these evaluations, the system classifies the job post and generates an analysis report containing the prediction result, company details, and timestamp information. The report can also be downloaded in PDF format for future reference. Overall, the use case diagram demonstrates how the system assists users in identifying fraudulent job postings and making safer job application decisions.

4.1.2 CLASS DIAGRAM

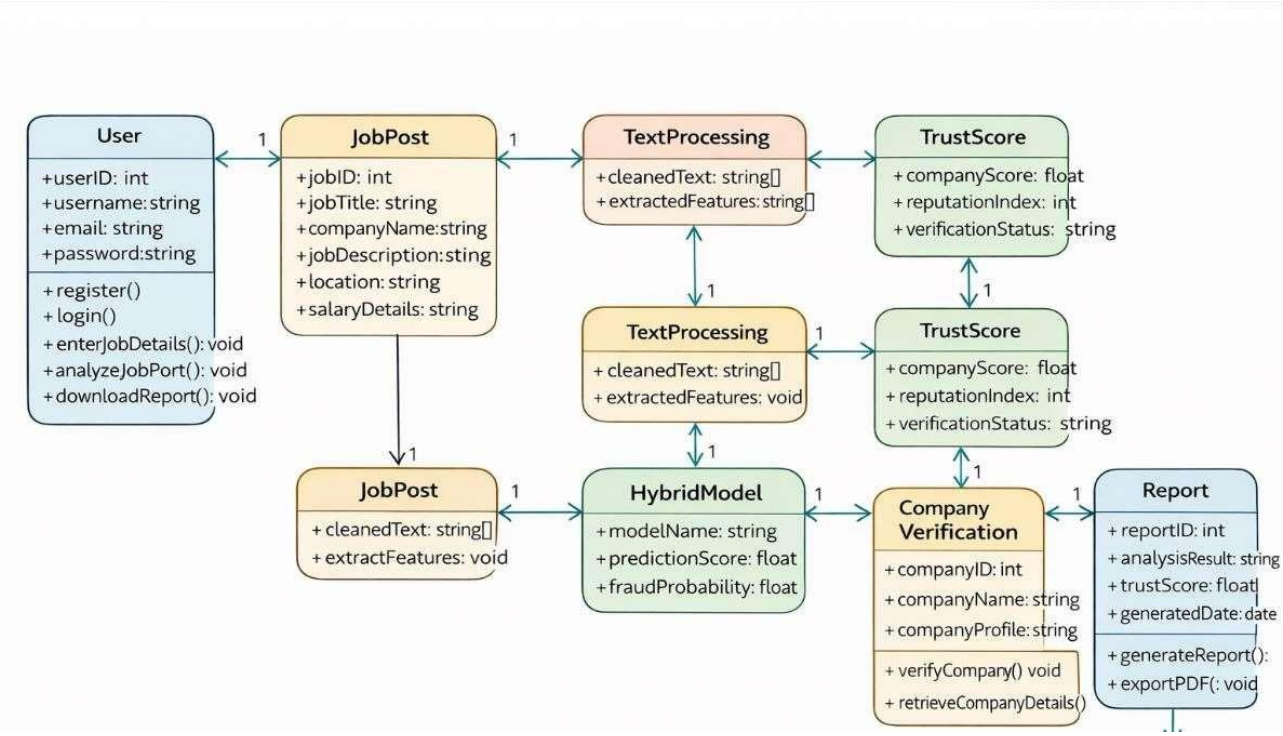


Fig. 4.1.2 Class Diagram for Recruitment Fraud Detection System

The Class Diagram represents the structural design of the Recruitment Fraud Detection System by illustrating the main classes and their relationships. The primary classes in this system include User, JobPost, TextProcessing, HybridModel, TrustScore, CompanyVerification, and Report modules. The User class is responsible for interacting with the system by submitting job

details and viewing analysis results. The JobPost class manages the job information provided by the user, including details such as job title, company name, job description, location, and salary information. The TextProcessing class prepares the job description by performing operations such as text cleaning, tokenization, and feature extraction. The HybridModel and TrustScore classes work together to analyse the job posting using machine learning techniques and evaluate the credibility of the company. The Report class generates the final analysis report containing the fraud prediction result and trust score. The Class Diagram helps clearly define the responsibilities and relationships between different modules of the system. Each class performs a specific function that contributes to detecting fraudulent recruitment postings. The interaction between classes ensures that job data flows smoothly from user input to the final analysis result. By organizing the system into separate classes, future modifications or improvements can be implemented easily without affecting the entire system. Overall, the Class Diagram provides a clear understanding of the internal structure of the recruitment fraud detection system and supports efficient system development.

4.1.3 SEQUENCE DIAGRAM

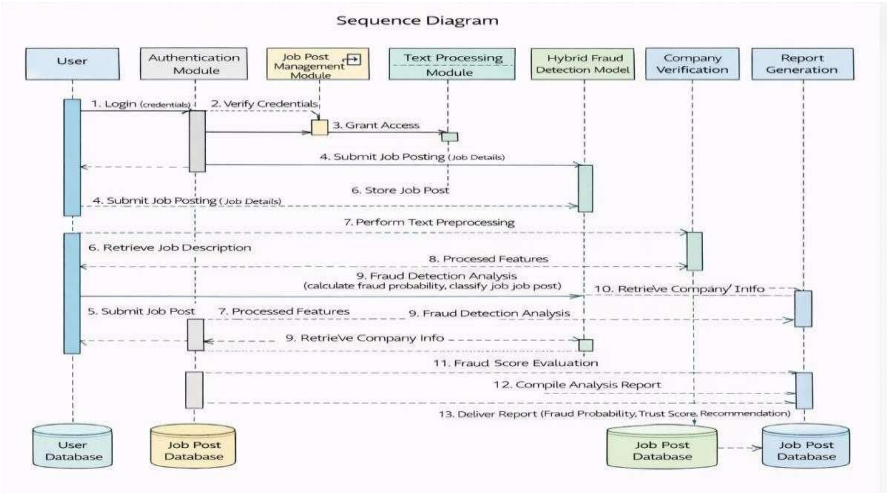


Fig 4.1.3 Sequence Diagram for Recruitment Fraud Detection System

The Sequence Diagram of the Recruitment Fraud Detection System illustrates the chronological interaction between the user, system interface, processing modules, and machine learning model during job analysis. The sequence begins when the user logs into the system and submits job posting details such as job title, company name, job description, location, and salary information through the application interface. The system then sends this information to the job management module, where the data is validated and stored for further processing. After validation, the job description is forwarded to the text processing module, where operations such as text cleaning, tokenization, and feature extraction are performed to prepare the data for analysis. The processed features are then passed to the hybrid fraud detection model, which evaluates suspicious patterns and calculates the fraud probability score. At the same time, the system performs company verification and trust score calculation to evaluate the credibility of the organization. The final results are then compiled into an analysis report and delivered to the user through the system interface. The Sequence Diagram helps clearly explain the order of interactions between system components. It shows how data flows between modules such as user input, text processing, fraud detection, company verification, and report generation. By representing these interactions step by step, the system workflow becomes easier to understand and implement.

4.1.4 ACTIVITY DIAGRAM

The Activity Diagram of the Recruitment Fraud Detection System illustrates the workflow of activities involved in analysing job postings to identify fraudulent recruitment advertisements. The process begins when the user accesses the system and enters job details such as job title, company name, job description, location, and salary information through the system interface. Once the information is submitted, the system performs text preprocessing operations such as cleaning the text, tokenizing the content, and normalizing the

data to prepare it for analysis. After preprocessing, the system extracts important textual features from the job description and sends them to the hybrid fraud detection model. The model analyses the features to identify suspicious patterns and calculates the fraud probability score. Based on this analysis, the system classifies the job posting as legitimate, suspicious, or fraudulent. The system then performs company verification and calculates a trust score to evaluate the credibility of the organization. Finally, the system generates an analysis report containing the prediction result, trust score, and related details, which is displayed to the user through the interface. The Activity Diagram helps clearly represent the flow of operations within the recruitment fraud detection system. It shows how each activity is connected from job submission to the final report generation. Overall, the diagram provides a clear understanding of the workflow involved in detecting fraudulent recruitment postings.

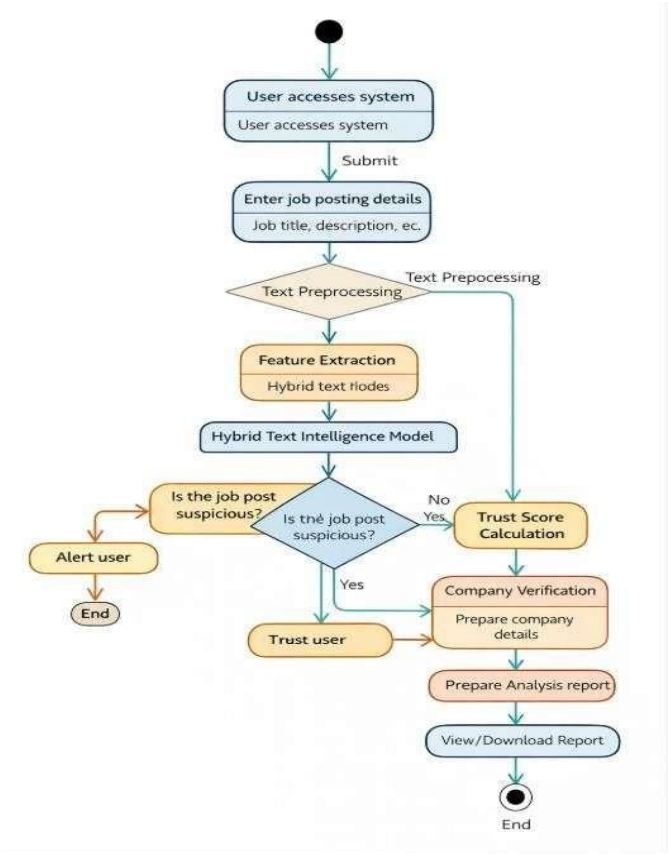


Fig 4.1.4 Activity Diagram for Recruitment Fraud Detection System

4.1.5 PACKAGE DIAGRAM

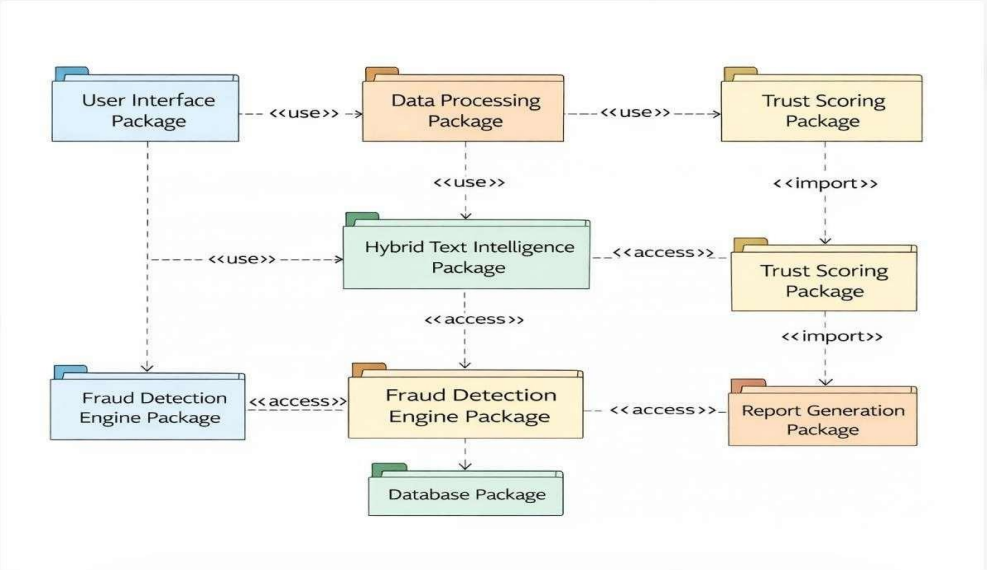


Fig 4.1.5 Package Diagram for Recruitment Fraud Detection System

The Package Diagram of the Recruitment Fraud Detection System represents the overall organization of the system by grouping related components into separate packages. It helps structure the system into different functional modules, making the design more organized and easier to understand. In this project, the system is divided into several packages such as the User Interface package, Data Processing package, Hybrid Text Intelligence package, Trust Scoring package, Fraud Detection Engine package, Report Generation package, and Database package. The User Interface package manages interactions between the user and the system, allowing users to enter job details and view analysis results. The Data Processing package handles operations such as text cleaning, tokenization, normalization, and feature extraction to prepare recruitment data for analysis. The Hybrid Text Intelligence and Fraud Detection packages analyse job postings using machine learning techniques and identify suspicious recruitment patterns, while the Trust Scoring package evaluates the credibility of the company or job post. The Report Generation package presents the final analysis result and allows the report to be downloaded, and the

Database package stores job posts, trust scores, and analysis records. This modular structure improves system maintainability and allows developers to manage different components efficiently. The Package Diagram helps organize system components into logical groups based on their functions. This separation improves clarity in system design and makes it easier to understand how different modules interact. It also supports better system maintenance since updates in one package can be made without affecting other modules. Overall, the Package Diagram provides a clear representation of the system architecture and helps manage the recruitment fraud detection system efficiently during development.

4.1.6 COLLABORATION DIAGRAM

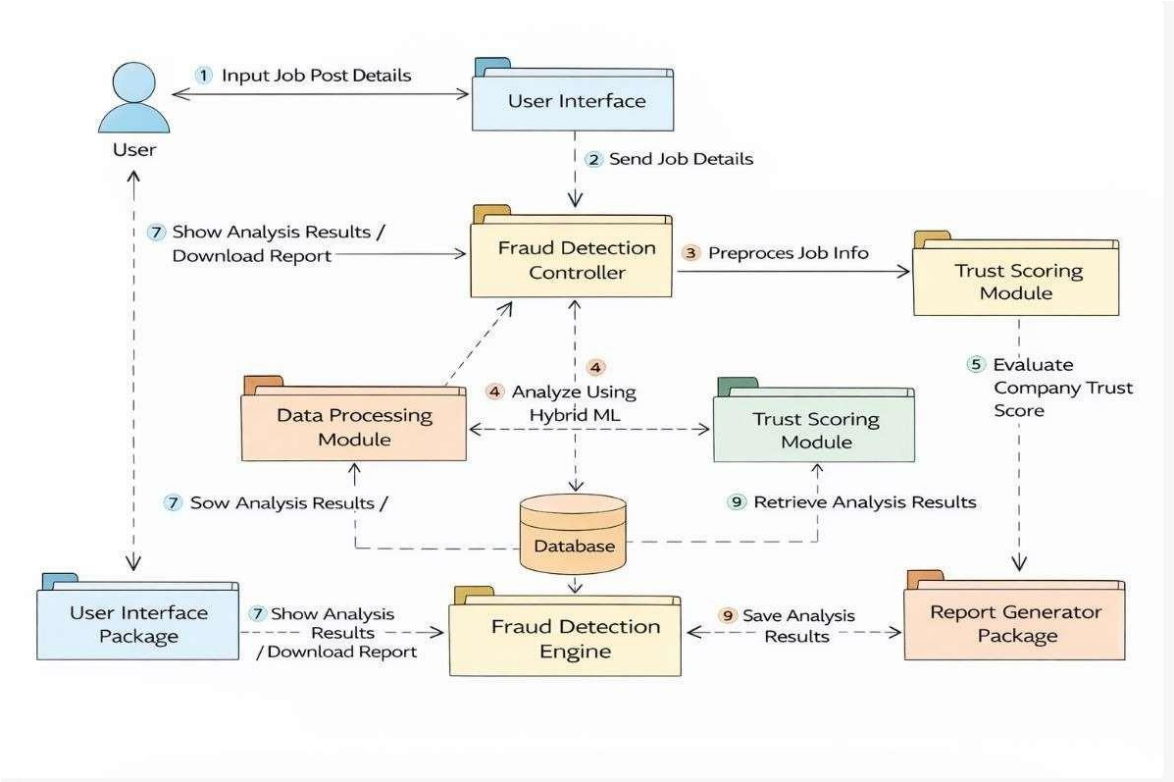


Fig 4.1.6 Collaboration Diagram for Recruitment Fraud Detection System

The Collaboration Diagram of the Recruitment Fraud Detection System represents how different components of the system work together to analyses job postings and detect fraudulent recruitment activities. It focuses on the

interaction between system modules and shows how they communicate to achieve the fraud detection process. In this project, the collaboration begins when the user enters job post details through the system interface. The interface sends the job information to the data processing module, where operations such as text cleaning, tokenization, normalization, and feature extraction are performed. After preprocessing, the processed data is forwarded to the hybrid text intelligence module for analysis. The system then evaluates suspicious patterns in the job description and calculates a fraud probability score. At the same time, the trust scoring module assesses the credibility of the company based on factors such as company details and historical records. Finally, the fraud detection engine combines the analysis results and trust score to classify the job post as legitimate, suspicious, or fraudulent. The result module then sends the final output back to the interface, where the analysis report is displayed to the user. The Collaboration Diagram helps in understanding how different modules coordinate and share information during the fraud detection process. It clearly shows the communication links between components such as the user interface, data processing module, fraud detection model, and report generation module. This representation improves the understanding of system interactions and ensures that each module performs its role effectively. Overall, the Collaboration Diagram highlights the cooperative behavior of system components in successfully identifying fraudulent recruitment postings.

4.2 DATA DICTIONARY

1. **JobPostID:**

   This attribute represents the unique identifier assigned to each job post submitted to the Recruitment Fraud Detection System. It helps the system track, store, and retrieve job postings during analysis and reporting.

2. **JobTitle:**

This feature indicates the title or designation mentioned in the recruitment post. It helps the system understand the nature of the job being offered and assists in analyzing patterns commonly associated with fraudulent job postings.

3. **CompanyName:**
   This attribute stores the name of the company associated with the job post. It is used to verify the authenticity of the organization and to evaluate the credibility of the recruitment source.
4. **JobDescription:**
   This feature contains the detailed textual information of the recruitment advertisement, including responsibilities, qualifications, salary information, and contact details. The job description is the primary input used for text analysis and fraud detection.
5. **ProcessedText:**
   This attribute represents the cleaned and preprocessed version of the job description. It is generated after applying text preprocessing techniques such as tokenization, stop-word removal, normalization, and feature extraction.
6. **FraudIndicators:**
   This feature stores the suspicious patterns or indicators identified during the hybrid text intelligence analysis. These indicators may include unrealistic salary offers, vague company information, urgent hiring language, or requests for payment.
7. **TrustScore:**
   This attribute represents the credibility score assigned to the job post based on company verification, textual analysis, and historical fraud patterns. The score helps determine whether the job post is legitimate or potentially fraudulent.
8. **DetectionResult:**
   This feature shows the final classification produced by the Fraud Detection Engine. The result categorizes the job posting as Legitimate, Suspicious, or Fraudulent based on the analysis and trust score.

9. **ReportID:**

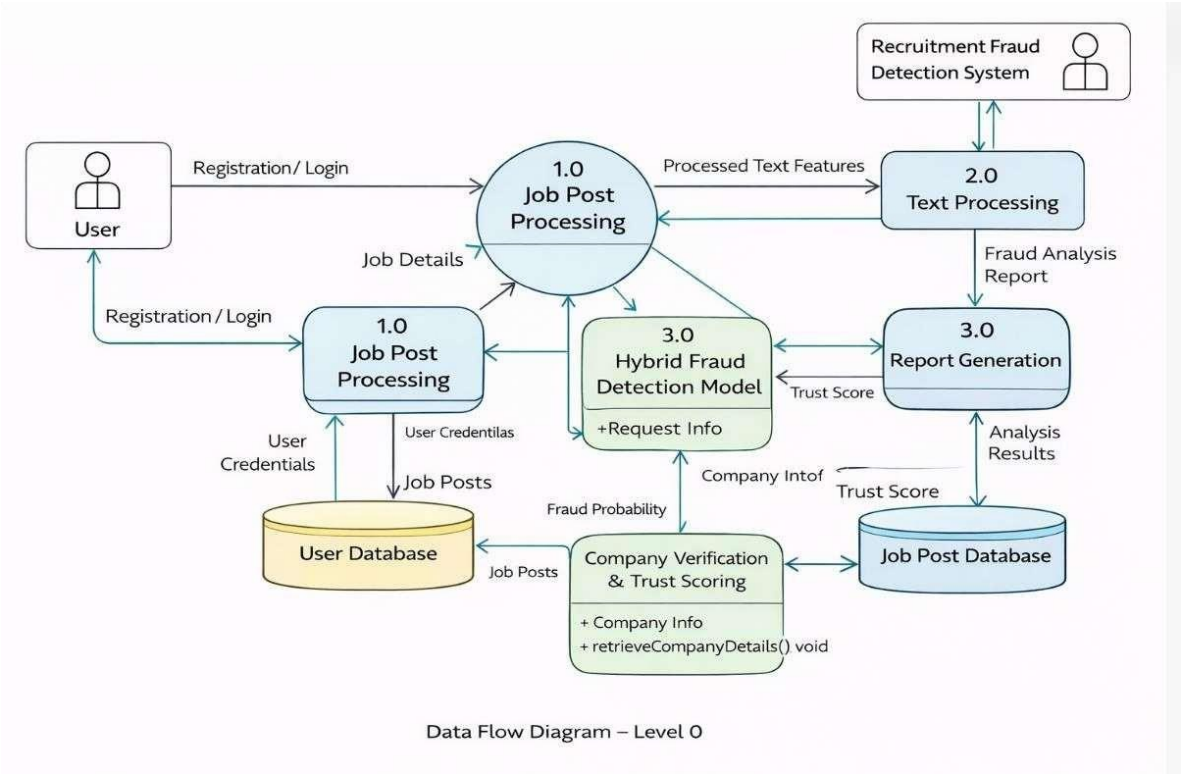
   This attribute represents the unique identifier assigned to the generated analysis report. It helps in managing and retrieving reports for user reference or auditing purposes.

10. **Timestamp:**

    This attribute records the date and time when the job post was analyzed and the fraud detection result was generated. It helps track system activity and maintain historical records of analyzed job posts.

4.3 DATAFLOW DIAGRAM

4.3.1 0 LEVEL DFD

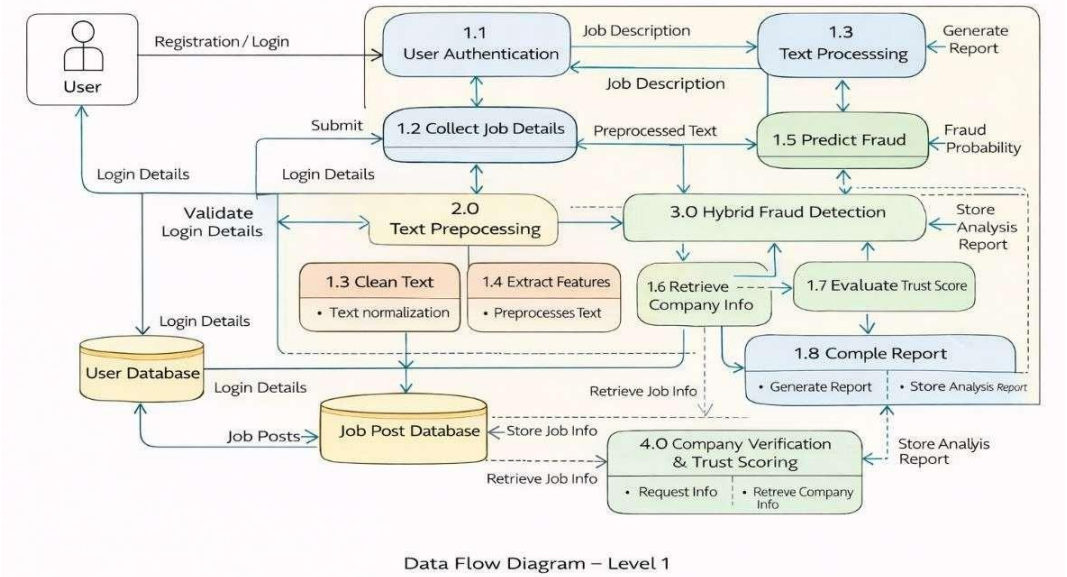


4.3.1 0 LEVEL DFD (CONTEXT DIAGRAM)

The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram,

represents the Recruitment Fraud Detection System as a single high-level process. It shows the overall interaction between the user and the system without describing the internal processing details. The user provides job posting information through the application interface for fraud analysis. The system processes the input data using text processing techniques, hybrid machine learning analysis, and trust scoring mechanisms to evaluate the authenticity of the job post. The output generated by the system is the fraud detection result along with a trust score and analysis report. The result may indicate whether the job posting is legitimate, suspicious, or fraudulent. This level mainly focuses on the overall data exchange between the user and the system. It helps define the system boundary and understand the main input and output flow. The Level 0 DFD provides a clear overview of how the recruitment fraud detection system operates.

4.3.2 1 LEVEL DFD

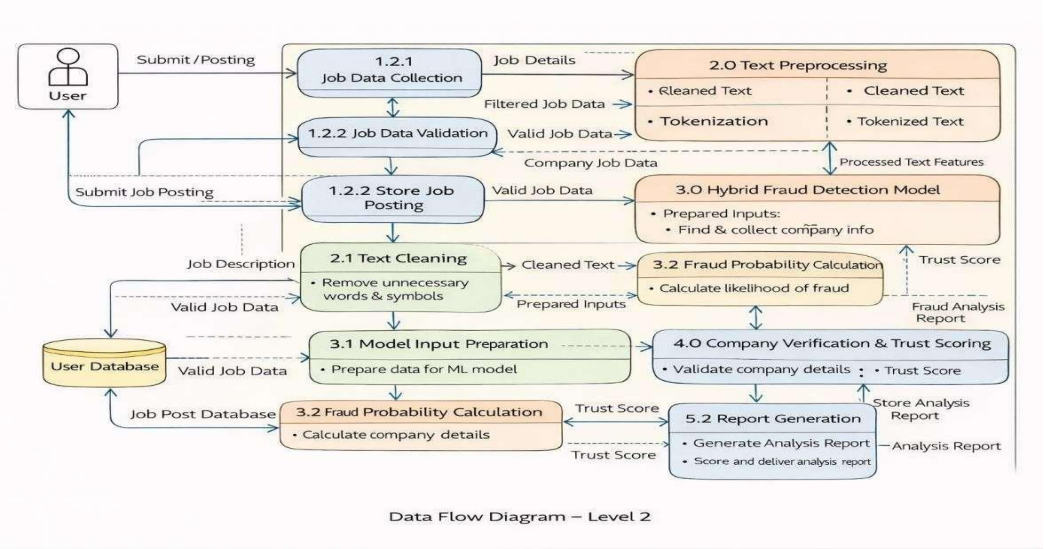


Data Flow Diagram – Level 1

The Level 1 Data Flow Diagram (DFD) provides a more detailed view of the Recruitment Fraud Detection System by dividing the main system into

several functional processes. It explains how job posting data moves through different modules before generating the final analysis result. At this level, the system is divided into processes such as User Authentication, Job Post Management, Text Preprocessing, Hybrid Fraud Detection, Trust Score Evaluation, and Report Generation. The process begins when the user logs into the system and submits job details such as job title, company name, job description, location, and salary information through the system interface. The job information is then stored and forwarded to the text preprocessing module where operations such as text cleaning, tokenization, and feature extraction are performed. After preprocessing, the processed data is passed to the hybrid fraud detection model where suspicious recruitment patterns are analyzed. At the same time, the system evaluates the credibility of the company through the trust scoring module. Finally, the system combines the fraud detection result and trust score to generate an analysis report, which is then displayed to the user through the application interface.

4.3.3 2 LEVEL DFD



4.3.3 2 LEVEL DFD for Recruitment Fraud Detection System

The Level 2 Data Flow Diagram (DFD) provides a detailed explanation of the internal processes involved in the Recruitment Fraud Detection System. It breaks down the processes shown in Level 1 into smaller steps to clearly represent how the system performs fraud detection for job postings. In this level, the process begins with job data collection, where details such as job title, company name, job description, location, and salary information are gathered from the user. The collected data then passes through a validation stage to ensure that the information is complete and properly formatted. After validation, the job description undergoes text preprocessing operations including text cleaning, tokenization, and feature extraction to convert the recruitment text into structured data. These extracted features are then analyzed by the hybrid fraud detection model, which identifies suspicious patterns and calculates the fraud probability score. At the same time, the system performs company verification and trust score calculation by evaluating company information and credibility indicators. Finally, the system combines the fraud detection result with the calculated trust score to generate an analysis report, which is then displayed to the user through the system interface.

4.4 ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship (ER) Diagram represents the structural workflow of the Recruitment Fraud Detection System by illustrating the main entities and their relationships involved in detecting fraudulent job postings. The process begins with the User, who submits job posting details through the system interface. These details are stored in the Job Post entity, which contains information such as job title, company name, job description, location, and salary details. The submitted job data is then processed by the Text Processing module, where operations such as text cleaning, tokenization, and feature extraction are performed to prepare the data for analysis. The processed features

are forwarded to the Fraud Analysis module, where the hybrid machine learning model evaluates suspicious patterns and determines whether the job post is legitimate or fraudulent. At the same time, the system performs Company Verification to evaluate the credibility of the organization associated with the job posting. Based on the verification results, the Trust Score module calculates the reliability score of the company. Finally, the analysis results and trust score are combined to generate a Report, which provides the final fraud detection result and recommendation to the user.

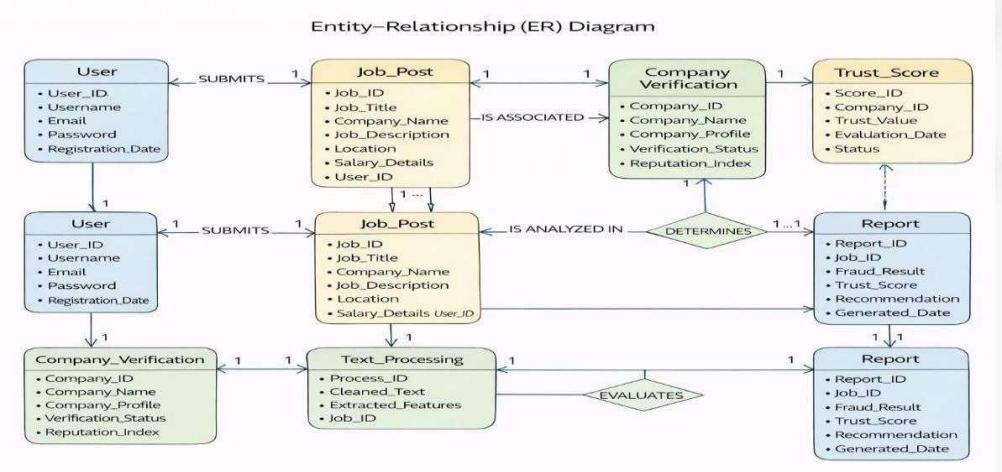


Fig 4.4 ENTITY RELATIONSHIP DIAGRAM for Recruitment Fraud Detection

Key Components of the ER Diagram:

- User: Initiates the process by submitting job posting details for analysis.
- Job Post: Represents the job advertisement entered by the user.
- Text Processing: Performs preprocessing tasks such as text cleaning and feature extraction.
- Fraud Analysis: Uses the hybrid machine learning model to evaluate fraudulent patterns.
- Company Verification: Validates the credibility of the company associated with the job post.
- Trust Score: Calculates the reliability score of the company.
- Report: Generates the final analysis result and recommendation for the user.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 ARCHITECTURE OVERVIEW

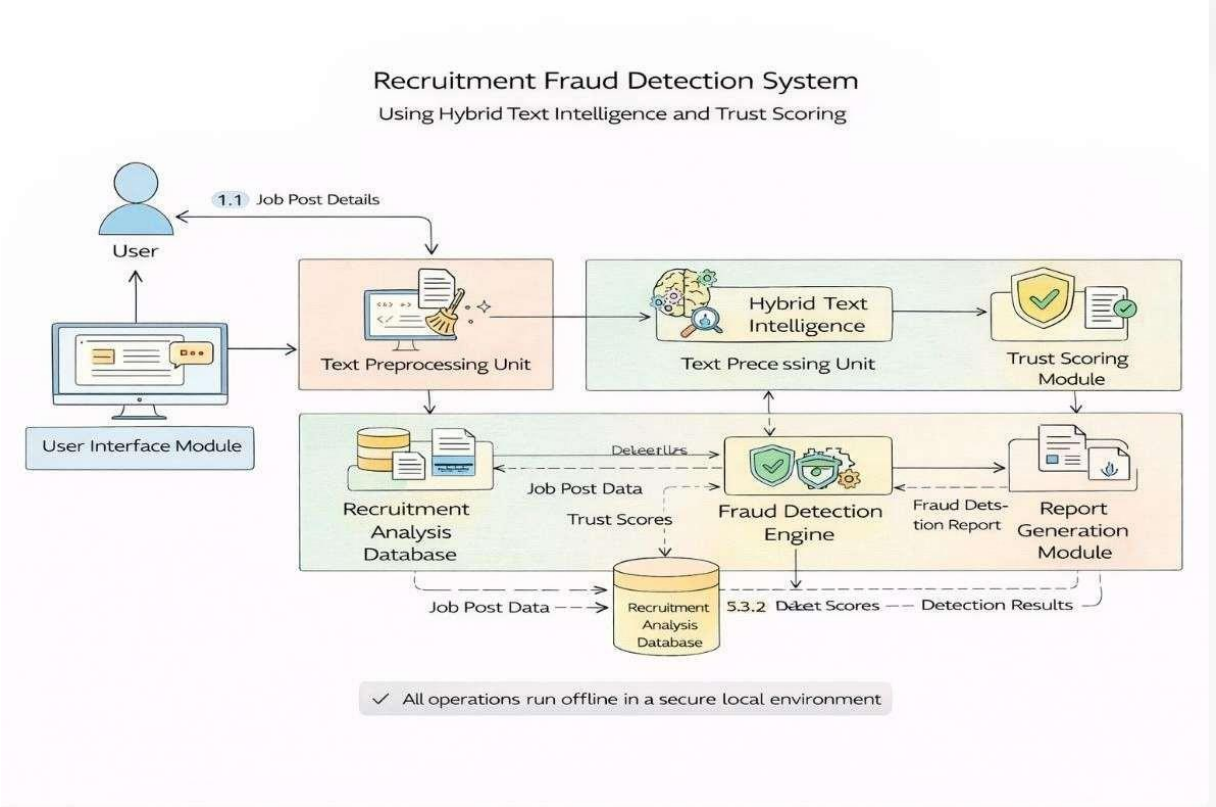


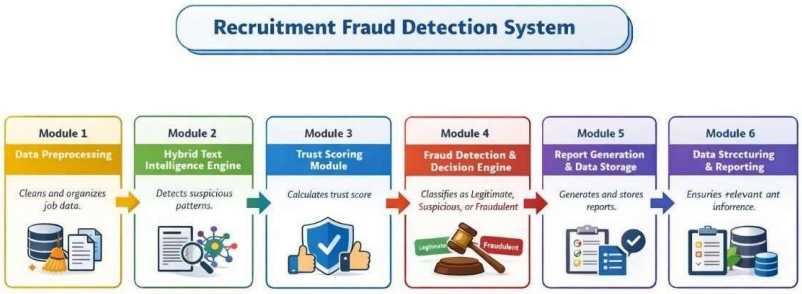
Figure 5.1 ARCHITECTURE OVERVIEW Recruitment Fraud Detection system

The architecture diagram illustrates the overall workflow of the Recruitment Fraud Detection System, which is designed to help users identify fraudulent job postings using hybrid text intelligence and trust scoring mechanisms. The process begins when the user enters job posting details such as job title, company name, job description, location, and salary information through the system interface. The user interface acts as the main communication layer between the user and the system, allowing users to easily submit job information and view the fraud analysis results. This interface is designed to be simple and accessible so that users can interact with the system efficiently. Once

the job details are submitted, the system collects the input data and transfers it to the text processing module. At this stage, several preprocessing operations are performed, including text cleaning, tokenization, and normalization to prepare the recruitment data for analysis. These preprocessing techniques ensure that the job description is properly structured and suitable for processing by the machine learning model.

After preprocessing, the extracted textual features are provided to the hybrid fraud detection module, which analyses the job posting using machine learning and text intelligence techniques. This module evaluates suspicious recruitment patterns and calculates the fraud probability score for the job post. At the same time, the system performs company verification and trust score evaluation by examining company information, credibility indicators, and available records. The outputs from the fraud detection model and trust scoring module are then combined to determine whether the job post is legitimate, suspicious, or fraudulent. Finally, the system generates a detailed analysis report containing the fraud prediction result, trust score, and recommendation. This report is displayed to the user through the system interface. By integrating modules such as user interface, text preprocessing, hybrid fraud detection, company verification, trust scoring, and report generation, the architecture provides a structured framework for accurate and efficient recruitment fraud detection. This organized workflow helps users quickly evaluate job postings and make safer employment decisions.

5.2 MODULE DESCRIPTION



5.2 MODULE DESCRIPTION Recruitment Fraud Detection system.

The recruitment fraud detection system is designed using a modular structure so that each stage of the workflow performs a specific task in analysing job postings. Dividing the system into modules improves clarity, maintainability, and efficiency during implementation. Each module processes a particular part of the data flow, starting from collecting job posting details to generating the final fraud analysis result. This structured approach ensures that the system can analyse large volumes of recruitment data while maintaining accuracy and reliability in identifying fraudulent job advertisements.

The modules work together in a sequential manner to transform raw job post information into meaningful fraud detection results. The process begins with user input and job data collection, followed by text preprocessing and feature extraction. After the data is properly prepared, the hybrid text intelligence engine and trust scoring module analyse the job posting to identify suspicious patterns and evaluate company credibility. Finally, the fraud detection engine combines these results and generates an analysis report for the user. This modular design allows easier system updates and improvements in future versions of the recruitment fraud detection system.

5.2.1 User Input & Data Collection Module

This module gathers all the recruitment information required for fraud analysis. In this project, the system collects job posting details such as job title, company name, salary information, job description, and contact details through the user interface. The main purpose of this module is to capture the raw recruitment data and organize it in a structured format for further processing. Once the information is received, it is validated and stored temporarily so that it can move to the preprocessing stage. This ensures that all job posting data enters the system in a consistent and usable format before text analysis and fraud detection begin.

5.2.2 Text Preprocessing & Feature Extraction Module

This is where the raw recruitment data gets ready for analysis. First, the system cleans the job posting text by removing unnecessary symbols, converting everything into a consistent format, and eliminating common stop words. Then it normalizes the content and breaks the text into smaller tokens so the system can understand the structure of the job description more clearly.

Next, the module organizes the processed text by identifying important words, phrases, and patterns within the job posting. These extracted features help highlight key elements such as salary information, company details, and job requirements. The main goal of this stage is to transform raw job post data into clean, structured information that can be effectively analysed by the fraud detection and trust scoring modules.

5.2.3 Hybrid Text Intelligence Engine Module

This module analyses the job posting content to identify patterns that may indicate recruitment fraud. It uses a combination of natural language processing and machine learning techniques to examine the cleaned and processed text

from the previous module. The system looks for suspicious indicators such as unrealistic salary offers, vague job requirements, urgent hiring language, or requests for registration fees.

At the same time, the module evaluates different textual features extracted from the job description to detect abnormal or misleading patterns commonly found in fraudulent recruitment posts. By combining multiple text analysis techniques, the Hybrid Text Intelligence Engine is able to capture both the meaning and structure of the job content. This helps the system identify potentially fraudulent recruitment advertisements with higher accuracy before moving to the trust scoring and decision stage.

5.2.4 Trust Scoring Module

This module evaluates the credibility of the job posting by analyzing the reliability of the recruitment source. It works alongside the text intelligence analysis by examining factors such as company name authenticity, contact information validity, job consistency, and historical fraud indicators stored in the system database. These factors help determine whether the recruitment source appears trustworthy or suspicious.

The module processes these parameters and calculates a trust score for each job posting. This score represents the overall reliability level of the recruitment advertisement. By combining credibility indicators with the textual analysis results from previous modules, the system gains a more balanced understanding of the job post. This helps improve the accuracy of fraud detection before the final classification stage.

5.2.5 Fraud Detection & Decision Engine Module

The Fraud Detection and Decision Engine module is responsible for analysing the results obtained from the previous stages of the recruitment fraud

detection system. It receives the outputs generated by the Hybrid Text Intelligence Engine and the Trust Scoring Module. These outputs include suspicious textual indicators from the job description and the credibility score calculated for the company or recruitment source. By combining these two sources of information, the system gains a more complete understanding of both the job content and the reliability of the organization offering the job.

This module evaluates various fraud indicators such as unrealistic salary offers, misleading job descriptions, incomplete company details, and unusual recruitment patterns. At the same time, it considers the trust score generated from company verification data and credibility checks. Based on this combined analysis, the module determines whether the job posting is legitimate, suspicious, or fraudulent. The final decision is then forwarded to the reporting module, where the results are stored and presented to the user as a fraud detection report. This module plays an important role in ensuring accurate and reliable identification of fraudulent recruitment advertisements.

5.2.6 Report Generation & Data Storage Module

The Report Generation and Data Storage module is responsible for presenting the final results of the recruitment fraud detection process in a clear and understandable format. After the Fraud Detection and Decision Engine completes the analysis, the system compiles the information into a structured report that includes details such as job title, company name, detected suspicious indicators, calculated trust score, and the final classification result. This output helps users understand whether the job posting appears legitimate, suspicious, or fraudulent and make informed decisions about the recruitment opportunity. The module also stores the processed job data and analysis results in the system database for future reference and review.

CHAPTER 6
SYSTEM IMPLEMENTATION

6.1 ALGORITHMS

6.1.1 Hybrid Text Intelligence and Trust Scoring Algorithm

The The Hybrid Text Intelligence and Trust Scoring algorithm is implemented in this project for the automatic detection and classification of fraudulent recruitment postings using textual job data. Online job portals contain thousands of job advertisements, and identifying fraudulent postings manually is both time-consuming and unreliable. Many recruitment scams appear legitimate at first glance and may mislead job seekers through attractive salary offers, vague company details, or misleading job descriptions. To overcome these challenges, the proposed system adopts an intelligent analytical approach that can examine job posting content and evaluate its authenticity.

The Hybrid Text Intelligence algorithm is designed to analyse the textual information contained in job advertisements such as job title, company name, salary details, job description, and contact information. By applying natural language processing techniques, the system can identify suspicious patterns and irregularities present in the recruitment data. Fraudulent job postings often contain indicators such as unrealistic salary promises, urgent hiring claims, incomplete company details, or requests for registration fees. The algorithm analyses these textual features to identify abnormal recruitment patterns.

Another important component of the system is the integration of a Trust Scoring mechanism. While textual analysis can detect suspicious patterns in job descriptions, evaluating the credibility of the recruitment source is equally important. The Trust Scoring model examines factors such as company legitimacy, email domain authenticity, contact information reliability, and historical recruitment data stored in the system database. Each of these

parameters contributes to a credibility score that reflects the overall trustworthiness of the job advertisement.

The Hybrid Fraud Detection framework therefore combines both textual analysis and credibility evaluation to provide accurate recruitment fraud identification. By integrating these two approaches, the system is capable of detecting hidden fraud indicators in job postings while also assessing the reliability of the recruitment source. This combined strategy improves the overall detection accuracy and helps job seekers avoid potential recruitment scams.

6.1.2 Working Mechanism of the Hybrid Fraud Detection Algorithm

The algorithm begins by receiving recruitment data submitted by the user, including job title, company name, salary information, job description, and contact details. These inputs form the primary dataset used for recruitment fraud analysis.

• The collected job posting information is then forwarded to the text preprocessing stage, where the system performs operations such as tokenization, stop-word removal, normalization, and noise reduction. These preprocessing steps ensure that the text data is cleaned and converted into a structured format suitable for analysis.

• After preprocessing, feature extraction techniques are applied to identify important textual characteristics present in the job advertisement. These features may include suspicious keywords, salary patterns, company information consistency, and structural patterns in the job description.

• The extracted textual features are then analyzed by the Hybrid Text Intelligence module, which evaluates the content of the job posting to detect suspicious recruitment patterns and abnormal language usage commonly associated with fraudulent job advertisements.

• At the same time, the Trust Scoring module evaluates the credibility of

the recruitment source by analysing parameters such as company authenticity, domain validity, and contact information reliability. Each factor contributes to the calculation of an overall trust score.

• Finally, the Fraud Detection Engine combines the suspicious textual indicators and the calculated trust score to classify the job posting as Legitimate, Suspicious, or Fraudulent, providing the final result to the user.

6.1.3 Text Preprocessing

Text preprocessing and feature extraction are important steps in preparing recruitment data for intelligent analysis. Job postings collected from online platforms often contain unstructured text, inconsistent formatting, and unnecessary symbols, which can reduce the effectiveness of machine learning algorithms if used directly. Therefore, the system performs preprocessing operations to clean and standardize the recruitment data before analysis.

During preprocessing, the job description text is converted into a normalized format by applying operations such as lower-casing, removal of special characters, and elimination of irrelevant symbols. Stop words that do not contribute meaningful information are also removed to reduce noise in the dataset. After this, tokenization is applied to divide the text into smaller units called tokens. Feature extraction techniques then identify important keywords, phrases, and patterns related to fraudulent recruitment behaviour. These features are converted into numerical representations so that they can be processed by the classification algorithm, improving the accuracy and efficiency of the fraud detection system.

6.1.4 Text Preprocessing and Feature Extraction Algorithm

The Trust Scoring Evaluation Model is designed to assess the credibility and reliability of the recruitment source. While textual analysis can

identify suspicious patterns in job descriptions, it is also important to evaluate whether the company or recruiter associated with the job posting appears legitimate. Fraudulent recruiters may manipulate job descriptions but often provide inconsistent company information or invalid contact details. Therefore, the trust scoring model adds an additional layer of verification in the recruitment fraud detection process.

The model evaluates several credibility parameters such as company name authenticity, domain verification of the recruiter's email address, availability of official company information, and consistency of contact details. Each parameter contributes to the overall credibility score of the job advertisement. Verified company details and official domain emails increase the trust score, while suspicious contact information or missing company data reduce it. The system combines these indicators to generate a final trust score representing the reliability of the recruitment source. This score is then used together with textual analysis results to improve the accuracy of recruitment fraud detection.

Trust Score Calculation Model

Let The trust score represents the reliability level of a recruitment advertisement. It is calculated by evaluating multiple credibility factors associated with the job posting. Each factor contributes a weighted value to the final trust score.

Let the credibility indicators be represented as:

- C_1 – Company authenticity score
- C_2 – Email domain validity score
- C_3 – Contact information reliability
- C_4 – Job information consistency
- C_5 – Historical recruitment reliability

Each indicator is assigned a weight depending on its importance. The final trust score T is calculated using the following expression:

$$T = w_1 C_1 + w_2 C_2 + w_3 C_3 + w_4 C_4 + w_5 C_5$$

Where:

- T represents the overall trust score of the job posting
- w_1, w_2, w_3, w_4, w_5 represent the weights assigned to each credibility factor
- C_1, C_2, C_3, C_4, C_5 represent the individual credibility scores

The trust score value typically ranges between 0 and 1, where higher values indicate greater credibility of the recruitment source.

Fraud Detection Classification Score

After calculating the trust score and analyzing textual features from the job description, the system combines both values to determine the final fraud detection result.

Let:

- F_t represent the textual fraud indicator score obtained from the Hybrid Text Intelligence module
- T represent the trust score calculated in the previous step

The final fraud detection score S is computed as:

$$S = \alpha F_t + (1 - \alpha)(1 - T)$$

Where:

- S represents the final fraud detection score
- α represents a balancing parameter between textual analysis and trust evaluation
- F_t represents suspicious textual pattern score
- T represents trust score of the job posting

Based on the calculated score S, the system classifies the recruitment advertisement into three categories:

- Legitimate – If S is below a predefined threshold

- Suspicious – If S falls within a moderate range
- Fraudulent – If S exceeds the fraud detection threshold

This combined scoring mechanism allows the system to evaluate both the content of the job posting and the credibility of the recruiter before making the final classification decision.

6.1.5 Integration of Hybrid Text Intelligence with Trust Scoring Framework

The integration of Hybrid Text Intelligence with the Trust Scoring framework forms the core architecture of the recruitment fraud detection system. Each component performs a specific role in analysing recruitment data, and their combined functionality enables the system to identify fraudulent job postings more reliably. The hybrid framework ensures that both textual content and recruiter credibility are evaluated together during the fraud detection process.

The first component focuses on textual analysis using natural language processing techniques. This module analyses job descriptions and identifies suspicious linguistic patterns, abnormal recruitment phrases, unrealistic salary offers, and misleading job requirements. These indicators help determine whether a job advertisement may contain fraudulent elements.

The second component evaluates the credibility of the recruitment source using the trust scoring model. By analysing company authenticity, domain validity, and contact information reliability, the system calculates a trust score representing the reliability of the job posting.

Once both analyses are completed, the results are combined within the fraud detection engine, which determines the final classification outcome. Based on this combined evaluation, the system categorizes the job posting as Legitimate, Suspicious, or Fraudulent, helping users identify potentially unsafe recruitment opportunities.

CHAPTER 7

SYSTEM TESTING

7.1 TESTCASES AND REPORTS

TEST CASE ID	TEST CASE / ACTION TO BE PERFORMED	EXPECTED RESULT	ACTUAL RESULT	PASS / FAIL
TC_01	Enter job posting details including job title, company name, salary, and description.	The system should accept the input data and display the job posting details for analysis.	The job posting information was accepted and displayed successfully.	Pass
TC_02	Submit the job analysis form with missing mandatory fields.	The system should display a validation message requesting the user to fill the required fields.	The system displayed a validation error and prevented submission.	Pass
TC_03	Submit a job posting containing realistic salary and verified company details.	The system should classify the job posting as Legitimate.	The system successfully classified the job as legitimate.	Pass
TC_04	Submit a job posting with unrealistic salary offers and suspicious contact information.	The system should detect suspicious patterns and classify the job as Fraudulent.	The system detected the suspicious indicators and marked the job as fraudulent.	Pass
TC_05	Submit a job	The system	The system	Pass

	posting with partially incomplete company information.	should classify the job posting as Suspicious.	categorized the job posting as suspicious.	
TC_06	Provide recruitment details with a verified company email domain.	The system should assign a high trust score for the job posting.	The system calculated a high trust score successfully.	Pass
TC_07	Provide recruitment details with an unverified email domain and inconsistent company information.	The system should assign a low trust score.	The system generated a low trust score correctly.	Pass
TC_08	Analyze multiple job postings sequentially.	The system should process all job postings and generate classification results without errors.	The system successfully analyzed all job postings and produced results.	Pass
TC_09	Generate a fraud detection report after job analysis.	The system should generate the fraud detection report.	The system generated the fraud detection report successfully.	Pass
TC_10	Store the analyzed job posting data in the system database.	The system should save the job data and analysis results for future reference.	The system successfully stored the job data and detection results.	Pass

Table 7.1 Test cases and reports for Recruitment Fraud Detection.

CHAPTER 8
CONCLUSION

8.1 CONCLUSION

The Recruitment Fraud Detection System using Hybrid Text Intelligence and Trust Scoring techniques has been successfully developed and implemented. The main objective of this project was to design an efficient system for automatically identifying fraudulent job postings using recruitment data. With the rapid growth of online job portals, recruitment fraud has become a serious concern for job seekers. Fraudulent job advertisements often mislead candidates by offering unrealistic salaries, vague job descriptions, or false company information. Traditional manual verification methods are often time-consuming and unreliable when a large number of job postings must be analysed. The proposed system addresses these challenges by applying intelligent data analysis techniques to detect suspicious recruitment patterns automatically.

The developed system integrates important components such as data preprocessing, feature extraction, trust score evaluation, and fraud classification. Text preprocessing techniques clean and normalize job posting information so that meaningful patterns can be extracted from recruitment data. Feature extraction methods analyse the textual content of job descriptions to identify suspicious keywords, abnormal recruitment phrases, and inconsistencies in job details, helping the system recognize patterns that may indicate fraudulent activities.

Another important component of the system is the Trust Scoring mechanism, which evaluates the credibility of the recruitment source. By analysing factors such as company authenticity, email domain validity, and reliability of contact information, the system calculates a trust score that reflects the legitimacy of the job posting. This trust score is combined with textual analysis results to provide a more accurate assessment of recruitment

authenticity.

Experimental evaluation shows that the proposed system can effectively identify recruitment fraud by classifying job postings as legitimate, suspicious, or fraudulent. The integrated hybrid framework improves detection accuracy by combining multiple analytical techniques within a single system. Overall, the system provides a practical solution for helping users verify job advertisements and supports a safer online recruitment environment.

8.2 FUTURE ENHANCEMENTS

Expansion to Multiple Recruitment Platforms

The proposed system is currently designed to analyze recruitment data entered by users or collected from limited job postings. In future work, the system can be expanded to automatically monitor and analyze job advertisements from multiple online recruitment platforms such as job portals, company websites, and social media recruitment pages. By integrating data from various platforms, the system will be able to detect fraudulent job postings on a larger scale. A multi-platform recruitment fraud detection system will provide better protection for job seekers and improve the reliability of online recruitment environments.

Integration with Real-Time Job Portal Monitoring

The system can be further enhanced by integrating it directly with job portals and recruitment websites. Through real-time monitoring, the system can automatically analyze newly posted job advertisements as soon as they appear online. By continuously scanning job postings and evaluating them using the fraud detection algorithm, suspicious job advertisements can be identified immediately. This real-time monitoring capability will help prevent fraudulent postings from reaching job seekers and will improve the safety of online recruitment platforms.

Mobile Application Development

The system can be improved by developing a mobile application that allows users to check the authenticity of job postings directly from their smartphones. Job seekers can enter or upload job details such as company name, job description, salary information, and contact details through the mobile application. The system will then analyze the information and instantly provide a fraud detection result. A mobile-based solution will make the system more accessible and convenient for users who frequently search for jobs through mobile devices.

Integration with AI-Based Company Verification

In the future, the system can incorporate advanced artificial intelligence techniques for company verification. The model can automatically cross-check company information with official company databases, business directories, and professional networking platforms. This verification process will help determine whether the company associated with the job posting is legitimate or potentially fraudulent. AI-based verification will significantly improve the accuracy of trust score evaluation and enhance the reliability of fraud detection.

Recommendation System for Safe Job Applications

In addition to identifying fraudulent job postings, the system can be enhanced to provide safety recommendations for job seekers. After analyzing a job advertisement, the system can suggest precautionary measures such as verifying company details, avoiding registration fee payments, or checking official recruitment channels. Providing safety guidelines will help job seekers make informed decisions when applying for jobs. Such a recommendation system will transform the fraud detection tool into a comprehensive job safety assistance platform.

APPENDICES

A.1 SUSTAINABLE DEVELOPMENT GOALS (SDG)

The proposed Recruitment Fraud Detection System contributes to sustainable development by enhancing transparency in the recruitment process, protecting job seekers from fraudulent practices, and promoting trust in digital employment platforms. By integrating hybrid text intelligence and trust scoring techniques, the system supports innovation in digital recruitment and fosters ethical, safe, and sustainable employment practices.

1. SDG 8: DECENT WORK AND ECONOMIC GROWTH

This goal promotes inclusive and safe employment opportunities. The system safeguards job seekers from fraudulent postings, ensuring access to legitimate job opportunities and supporting fair labor practices, which contribute to sustained economic growth.

2. SDG 16: PEACE, JUSTICE AND STRONG INSTITUTIONS

By detecting and flagging fraudulent recruitment activity, the system promotes transparency and accountability in the employment sector. This strengthens institutional integrity and protects individuals from exploitation.

3. SDG 9: INDUSTRY, INNOVATION AND INFRASTRUCTURE

The system leverages AI, natural language processing, and trust scoring algorithms to innovate recruitment verification. This demonstrates the application of intelligent technology in strengthening digital infrastructure for safer job marketplaces.

4. SDG 4: QUALITY EDUCATION

By informing job seekers about potential recruitment scams, the system educates and empowers users to make informed career decisions. This contributes to lifelong learning and digital literacy in employment practices.

A.2 SOURCE CODE

A.2.1 Python Backend Code

```
import pandas as pd
import nltk
import re
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
df = pd.read_csv("fake_job_postings.csv")
df.head()
# Handle missing values
text_columns = ['title', 'company_profile', 'requirements', 'benefits', 'location']
for col in text_columns:
    if col in df.columns:
        df[col] = df[col].fillna("")

# Remove duplicates
df.drop_duplicates(inplace=True)

# Check label distribution
df['fraudulent'].value_counts()
df['description'] = df['description'].fillna("")
stop_words = set(stopwords.words('english'))
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()

def clean_text_lemma(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [lemmatizer.lemmatize(w) for w in tokens if w not in stop_words]
    return " ".join(tokens)

df['clean_description'] = df['description'].apply(clean_text_lemma)
```

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [w for w in tokens if w not in stop_words]
    return " ".join(tokens)
df['clean_description'] = df['description'].apply(clean_text)
df[['description', 'clean_description']].head()
X = df['clean_description']
y = df['fraudulent']
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    max_features=8000,
    ngram_range=(1,2)
)

X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
from sklearn.naive_bayes import MultinomialNB

model = MultinomialNB()
model.fit(X_train_vec, y_train)
from sklearn.metrics import accuracy_score, classification_report

y_pred = model.predict(X_test_vec)
prediction = best_model.predict(sample_vec)[0]
confidence = best_model.predict_proba(sample_vec).max()
```

```
print("Prediction:", "Fake" if prediction == 1 else "Genuine")
print("Confidence:", round(confidence * 100, 2), "%")
feature_names = vectorizer.get_feature_names_out()
top_features = best_model.feature_log_prob_[1].argsort()[-10:]

print("Top suspicious keywords:")
for i in top_features:
    print(feature_names[i])
from urllib.parse import urlparse
import requests
def check_website_security(url):
    try:
        parsed = urlparse(url)

        if parsed.scheme == "https":
            return "Secure (HTTPS)"
        elif parsed.scheme == "http":
            return "Not Secure (HTTP)"
        else:
            return "Invalid URL"

    except:
        return "Invalid URL"
print(check_website_security("https://careers.google.com"))
print(check_website_security("http://example-job-site.com"))
print(check_website_security("xyz"))
df['text_length'] = df['clean_description'].apply(len)
def caps_ratio(text):
    if len(text) == 0:
        return 0
    return sum(1 for c in text if c.isupper()) / len(text)

df['caps_ratio'] = df['description'].apply(caps_ratio)
suspicious_keywords = [
```

```
    "earn", "no experience", "work from home",
    "urgent", "immediate", "fees", "registration",
    "whatsapp", "telegram"
]

def suspicious_count(text):
    text = text.lower()
    return sum(1 for kw in suspicious_keywords if kw in text)

df['suspicious_count'] = df['description'].apply(suspicious_count)
from scipy.sparse import hstack
from sklearn.preprocessing import StandardScaler
import numpy as np

numeric_features = df[['text_length', 'caps_ratio', 'suspicious_count']]

scaler = StandardScaler()
numeric_scaled = scaler.fit_transform(numeric_features)

numeric_scaled_sparse = np.hstack(numeric_scaled)
X_train_combined = hstack([X_train_vec, numeric_scaled[:len(X_train)]])

for i in range(5):
    print("Risk:", risk_level(probs[i][1]),
          "| Confidence:", round(probs[i][1]*100,2), "%")
def website_score(url):
    if url.startswith("https"):
        return 1
    elif url.startswith("http"):
        return 0.5
    else:
        return 0
final_score = 0.7 * probs[0][1] + 0.3 * website_score("https://example.com")
```

```
print("Final Risk Level:", risk_level(final_score))
print("Final Confidence:", round(final_score*100,2), "%")
suspicious_keywords = [
    "work from home", "earn money", "no experience",
    "easy money", "quick money", "limited slots",
    "urgent hiring", "immediate joining",
    "registration fee", "processing fee"
]
def count_suspicious_keywords(text):
    text = text.lower()
    return sum(1 for kw in suspicious_keywords if kw in text)

df['suspicious_keyword_count'] = df['clean_description'].apply(count_suspicious_keywords)
def count_exclamation_marks(text):
    return text.count('!')

def urgency_flag(text):
    text = text.lower()
    return int(any(p in text for p in urgency_phrases))

df['urgency_flag'] = df['clean_description'].apply(urgency_flag)
df.head()
def punctuation_count(text):
    return text.count('!') + text.count('?')

df['punctuation_count'] = df['description'].fillna("").apply(punctuation_count)
def readability_score(text):
    words = text.split()
    if len(words) == 0:
        return 0
    avg_word_length = sum(len(w) for w in words) / len(words)
    return avg_word_length

df['readability_score'] = df['clean_description'].apply(readability_score)
```

```
df['title_length'] = df['title'].fillna("").apply(lambda x: len(x.split()))
df['title_desc_ratio'] = df['title_length'] / (df['description_length'] + 1)
urgency_phrases = [
    "urgent", "immediate", "join immediately",
    "hiring now", "limited time"
]

def urgency_flag(text):
    text = text.lower()
    return int(any(p in text for p in urgency_phrases))

df['urgency_flag'] = df['clean_description'].apply(urgency_flag)
from sklearn.feature_extraction.text import TfidfVectorizer
tfidf = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1,2),
    stop_words='english'
)
X = tfidf.fit_transform(df['clean_description'])
X_text = tfidf.fit_transform(df['clean_description'])
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

y = df['fraudulent']   # or your label column

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
# TF-IDF
tfidf = TfidfVectorizer(max_features=5000)
X_text = tfidf.fit_transform(df["clean_description"])
```

```
# Train model
lr_text = LogisticRegression(max_iter=1000)
lr_text.fit(X_text, y)
# Save both together
from joblib import dump
dump(lr_text, "fake_job_lr_model.pkl")
dump(tfidf, "tfidf_vectorizer.pkl")
feature_cols = [
    'description_length',
    'caps_ratio',
    'suspicious_keyword_count',
    'punctuation_count',
    'readability_score',
    'title_desc_ratio',
    'urgency_flag'
]
X_meta = df[feature_cols]
X_meta.head()
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_meta_scaled = scaler.fit_transform(X_meta)
from scipy.sparse import hstack

X_final = hstack([X_text, X_meta_scaled])
X_final.shape
from sklearn.model_selection import train_test_split

y = df['fraudulent']   # label column

X_train, X_test, y_train, y_test = train_test_split(
    X_final, y, test_size=0.2, random_state=42, stratify=y
)
X_train.shape, X_test.shape
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import classification_report, accuracy_score
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))
print(classification_report(y_test, y_pred_lr))
# OPTIONAL: Naive Bayes using TF-IDF ONLY (no metadata)
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report, accuracy_score

X_text_train, X_text_test, y_train_nb, y_test_nb = train_test_split(
    X_text, y, test_size=0.2, random_state=42, stratify=y
)
nb = MultinomialNB()
nb.fit(X_text_train, y_train_nb)

y_pred_nb = nb.predict(X_text_test)

print("Naive Bayes (TF-IDF only) Accuracy:", accuracy_score(y_test_nb, y_pred_nb))
print(classification_report(y_test_nb, y_pred_nb))
# ===== PHASE 5 (FIX): TEXT-ONLY MODEL =====

# 1. TF-IDF on cleaned text
X_text = tfidf.fit_transform(df['clean_description'])
y = df['fraudulent']
# 2. Train-test split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X_text, y, test_size=0.2, random_state=42
)
# 3. Train Logistic Regression (TEXT ONLY)
from sklearn.linear_model import LogisticRegression
lr_text = LogisticRegression(max_iter=1000)
```

```
lr_text.fit(X_train,  y_train)

print("Text-only Logistic Regression trained successfully")
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
    n_jobs=-1
)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
import numpy as np

# Convert binary labels to 3-class for training
# 0 → Real
# 1 → Suspicious
# 2 → Fake

def create_multiclass_label(row):
    if row == 0:
        return 0
    else:
        return 2
df['multi_label'] = df['fraudulent'].apply(create_multiclass_label)
y_multi = df['multi_label']

X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(
    X_final, y_multi, test_size=0.2, random_state=42, stratify=y_multi
)
lr_multi = LogisticRegression(max_iter=1000)
lr_multi.fit(X_train_m, y_train_m)
```

```
y_pred_multi = lr_multi.predict(X_test_m)
print(classification_report(y_test_m, y_pred_multi))
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

tfidf = TfidfVectorizer(
    max_features=5007,  # ← VERY IMPORTANT
    ngram_range=(1,2),
    stop_words="english"
)
X = tfidf.fit_transform(df["description"])  # or similar column
y = df["fraudulent"]

lr_text = LogisticRegression(max_iter=1000)
lr_text.fit(X, y)
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
import joblib

# Correct columns (adjust names if needed)
texts = df["description"]        # or df["text"]
labels = df["fraudulent"]        # 0 = legit, 1 = fraud

tfidf = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1,2),
    stop_words="english"
)

X = tfidf.fit_transform(texts)
y = labels

lr_text = LogisticRegression(max_iter=1000)
lr_text.fit(X, y)
```

```
joblib.dump(tfidf, "tfidf_vectorizer.pkl")
joblib.dump(lr_text, "fake_job_lr_model.pkl")

print("■□ Clean text-only model saved")
import joblib

joblib.dump(tfidf, "tfidf_vectorizer.pkl")
joblib.dump(lr_text, "fake_job_lr_model.pkl")

print("■□ Model & vectorizer saved together")
import joblib

joblib.dump(lr,   "fake_job_lr_model.pkl")
joblib.dump(tfidf,    "tfidf_vectorizer.pkl")
joblib.dump(scaler,    "feature_scaler.pkl")
def map_risk(prob):
    """
    Convert fraud probability into risk category
    """
    if prob < 0.3:
        return "Legitimate"
    elif prob < 0.7:
        return "Suspicious"
    else:
        return "Fraudulent"
def predict_job_risk(
    raw_text,
    model,
    vectorizer,
    scaler,
    feature_columns
):
    """
```

```
    Full pipeline prediction:
    Input  -> raw job description
    Output -> risk label + confidence
    """
    # ---- Text cleaning (reuse your Phase 2 logic) ----
    cleaned = clean_text_lemma(raw_text)

    # ---- TF-IDF ----
    text_vec = vectorizer.transform([cleaned])

    # ---- Feature engineering (Phase 3 features) ----
    features = {}

    features['description_length'] = len(raw_text)
    features['caps_ratio'] = sum(1 for c in raw_text if c.isupper()) / max(len(raw_text), 1)

suspicious_words = [
        "earn", "quick money", "no experience",
        "work from home", "urgent", "guaranteed"
    ]
    features['suspicious_keyword_count'] = sum(
        raw_text.lower().count(w) for w in suspicious_words
    )

    features['punctuation_count'] = raw_text.count("!") + raw_text.count("$")
    features['readability_score'] = 206.835 - 1.015 * len(raw_text.split())
    features['title_desc_ratio'] = 1

    features['urgency_flag'] = int(any(w in raw_text.lower() for w in ["urgent", "immediate"]))
    meta_df = pd.DataFrame([features])[feature_columns]
    meta_scaled = scaler.transform(meta_df)
```

A.2.2 APP.PY

```
import import streamlit as st
import joblib
import pandas as pd
import matplotlib.pyplot as plt
from datetime import date
from reportlab.lib.pagesizes import A4
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib import colors as rl_colors
import os
from hybrid_engine import hybrid_decision_engine
from database import get_prediction_stats, fetch_all_results
# ---------- PAGE CONFIG ----------
st.set_page_config(
    page_title="Fake Job Post Detection",
    layout="wide"
)
# ---------- GLOBAL STYLES ----------
st.markdown("""
<style>
/* ================ PASTEL CLOUD BACKGROUND ================ */
.stApp {
    background: linear-gradient(
        160deg,
        #e9d5ff 0%,  /* light lavender */
        #dbeafe 25%,  /* soft blue */
        #f5d0fe 50%,  /* pastel purple */
        #c7d2fe 75%,  /* bluish violet */
        #fbcfe8 100%  /* light pink */
    );
    background-attachment: fixed;
}
/* ================ GLOBAL CLEANUP ================ */
```

```
header { visibility: hidden; }

.block-container {
    padding-top: 2rem;
}
/* ================= GLASS CARD ================= */
.glass-card {
    background: rgba(17, 24, 39, 0.85);
    border-radius: 20px;
    padding: 28px;
    box-shadow: 0 12px 30px rgba(0,0,0,0.35);
}
/* ================= TEXT ================= */
.center {
    text-align: center;
    color: white;
}

.feature-list {
    max-width: 600px;
    margin: auto;
    color: #111827;
    font-size: 15px;
}

.feature-list li {
    margin-bottom: 8px;
}
/* ================= REMOVE BLACK BARS ================= */
div[data-testid="column"] {
    padding-top: 0rem !important;
    padding-bottom: 0rem !important;
}
.stButton > button {
```

```
    margin-top: 0px !important;
    margin-bottom: 0px !important;
}
hr {
    display: none;
}
</style>
""", unsafe_allow_html=True)
def interpret_risk(score):
    if score >= 0.75:
        return "🔴 High Risk – Strong fraud indicators detected"
    elif score >= 0.45:
        return "☐ Medium Risk – Suspicious patterns found"
    else:
        return "☐ Low Risk – Job post appears legitimate"

def generate_pdf_report(recent_df,  stats):
    file_name = f"job_detection_report_{date.today()}.pdf"
    file_path = os.path.join(os.getcwd(), file_name)

    doc = SimpleDocTemplate(file_path, pagesize=A4)
    styles = getSampleStyleSheet()
    elements = []
    # Title
    elements.append(Paragraph(
        "<b>Fake Job Detection Analytics Report</b>",
        styles["Title"]
    ))
    elements.append(Spacer(1, 12))
    # Date
    elements.append(Paragraph(
        f"Generated on: {date.today().strftime('%d-%m-%Y')}",
        styles["Normal"]
    ))
```

```
elements.append(Spacer(1, 12))
# Summary
elements.append(Paragraph("<b>Prediction Summary</b>", styles["Heading2"]))
for label, count in stats:
    total = sum(c for _, c in stats)
    elements.append(
        Paragraph(f"{label}: {count} ({(count/total)*100:.1f}%)", styles["Normal"])
    )
elements.append(Spacer(1, 15))
elements.append(Paragraph(
    "<b>Analyst Note</b><br/>"
    "Fraudulent job posts commonly exhibit urgency, unverifiable domains, "
    "and non-corporate email addresses. Users are advised to verify company "
    "authenticity before applying.",
    styles["Normal"]
))
elements.append(Spacer(1, 12))
elements.append(Paragraph("<b>Overall Risk Distribution</b>", styles["Heading2"]))
summary_table = Table(
    [["Risk Type", "Count"]] + [[label, count] for label, count in stats]
)
summary_table.setStyle(TableStyle([
    ("BACKGROUND", (0,0), (-1,0), rl_colors.HexColor("#4F46E5")),
    ("TEXTCOLOR", (0,0), (-1,0), rl_colors.white),
    ("GRID", (0,0), (-1,-1), 1, rl_colors.black),
    ("ALIGN", (1,1), (-1,-1), "CENTER"),
    ("FONT", (0,0), (-1,0), "Helvetica-Bold")
]))
elements.append(summary_table)

# Recent Results Table
elements.append(Paragraph("<b>Recent Job Detections</b>", styles["Heading2"]))
table_data = [recent_df.columns.tolist()] + recent_df.values.tolist()
```

```
    table = Table(table_data, repeatRows=1)
    table.setStyle(TableStyle([
        ("BACKGROUND", (0,0), (-1,0), rl_colors.lightgrey),
        ("GRID", (0,0), (-1,-1), 1, rl_colors.black),
        ("FONT", (0,0), (-1,0), "Helvetica-Bold"),
        ("ALIGN", (1,1), (-1,-1), "CENTER")
    ]))
    elements.append(table)
    elements.append(Spacer(1, 30))
    elements.append(Paragraph(
        "Generated by Fake Job Post Detection System | Final Year Project",
        styles["Italic"]
    ))

    doc.build(elements)

    return file_path
# ---------- LOAD MODEL ----------
vectorizer = joblib.load("vectorizer.pkl")
model = joblib.load("model.pkl")

# ---------- SESSION STATE ----------
if "page" not in st.session_state:
    st.session_state.page = "Home"

# ---------- TOP NAV ----------
nav1, nav2, nav3 = st.columns(3)
with nav1:
    if st.button("🏠 Home"):
        st.session_state.page = "Home"
with nav2:
    if st.button("🔍 Detection"):
        st.session_state.page = "Detection"
with nav3:
```

```
    if st.button("📊 Analytics"):
        st.session_state.page = "Analytics"

st.markdown("---")

# ====================================================
# 🏠HOME PAGE
# ====================================================
if st.session_state.page == "Home":

    st.markdown("""
    <div class="glass-card center">
        <h1>🕵️ Fake Job Post Detection System</h1>
        <p style="color:#e5e7eb;">
            Hybrid AI-based framework to detect fraudulent recruitment posts
        </p>
    </div>
    """, unsafe_allow_html=True)
    st.markdown("<br>", unsafe_allow_html=True)
    col1, col2, col3 = st.columns([1,2,1])
    with col2:
        st.markdown("""
        <div class="glass-card center">
            <h3>⚡ About the Project</h3>
            <p  style="color:#d1d5db;font-size:15px;">
                Online recruitment platforms are increasingly targeted by fraudsters.
                This system automatically identifies fake job postings using
                NLP, scam-pattern detection, and company trust verification.
            </p>
        </div>
        """, unsafe_allow_html=True)

    st.markdown("<br><br>", unsafe_allow_html=True)
```

```
    st.markdown("""
    <h3 style="text-align:center;">Key Features</h3>
    <ul class="feature-list">
        <li>Hybrid Machine Learning + Rule-based fraud detection</li>
        <li>Advanced NLP-based job text analysis</li>
        <li>Company trust and domain verification</li>
        <li>Explainable fraud risk scoring</li>
        <li>Interactive analytics dashboard</li>
    </ul>
    """, unsafe_allow_html=True)

# ========================================================
# DETECTION PAGE
# ========================================================
elif st.session_state.page == "Detection":

    st.markdown("<h2 class='center'> Job Post Analysis</h2>", unsafe_allow_html=True)
    st.markdown("<br>", unsafe_allow_html=True)

    col_input, col_result = st.columns([1.1, 0.9])

    # ---------- INPUT ----------
    with col_input:
        st.markdown("<div class='glass-card'>", unsafe_allow_html=True)

        job_text = st.text_area(" Job Description", height=180)
        company_name = st.text_input(" Company Name")
        website_url = st.text_input(" Company Website")
        email = st.text_input(" Contact Email")
        location = st.text_input(" Job Location")

        analyze = st.button(" Analyze Job Post")
        st.markdown("</div>", unsafe_allow_html=True)
```

```
# ---------- RESULT ----------
with col_result:
    if analyze:
        if not job_text or not company_name:
            st.warning("Please fill Job Description and Company Name") else:
            result = hybrid_decision_engine(
                job_text, company_name, website_url,
                model, vectorizer, email, location
            )
            st.markdown("<div class='glass-card'>", unsafe_allow_html=True)
            label = result["risk_label"]
            score = result["final_score"]
            if label == "Legitimate":
                st.markdown("<br>", unsafe_allow_html=True)
                st.markdown("<h3 style='color:#22c55e;'>🏢 Company Profile</h3>",
unsafe_allow_html=True)
                profile = fetch_company_profile(company_name)
                if profile:
                    st.markdown(f"""<p
style="color:#111827;"><b>Overview:</b><br>{profile.get("description", "No public
description available.")}</p>""", unsafe_allow_html=True)

                    if profile.get("source"):
                        st.markdown(f"<a href='{profile['source']}' target='_blank'>🔗 View public
source</a>",unsafe_allow_html=True)

                    else:
                        st.info("Public company details not available.")

    # ================ REAL-TIME JOB LINKS ================
                    st.markdown("<hr>", unsafe_allow_html=True)
                    st.markdown("<h3>🔗 Cross-Platform Job Search</h3>",
```

```
unsafe_allow_html=True)
                query = f"{company_name} {location} jobs"
                platforms = {"Google Jobs": f"https://www.google.com/search?q={query}",
                        "LinkedIn":
f"https://www.linkedin.com/jobs/search/?keywords={query}",
                        "Indeed": f"https://www.indeed.com/jobs?q={query}",
                        "Naukri": f"https://www.naukri.com/{query.replace(' ', '-')}-jobs"
                        }
                for name, link in platforms.items():
                   st.markdown(f"🔷 <a href='{link}' target='_blank'>{name}</a>",
unsafe_allow_html=True)

            if label == "Fraudulent":
               st.markdown("<h2 style='color:#ff4b4b;'>🚨 FRAUDULENT JOB
POST</h2>", unsafe_allow_html=True)
            elif label == "Suspicious":
               st.markdown("<h2 style='color:#facc15;'>⚠️ SUSPICIOUS JOB POST</h2>",
unsafe_allow_html=True)
            else:
               st.markdown("<h2 style='color:#22c55e;'>✅ LEGITIMATE JOB POST</h2>",
unsafe_allow_html=True)

            c1, c2, c3 = st.columns(3)
            c1.metric("Final Risk", round(score, 2))
            c2.metric("Text Risk", round(result["text_score"], 2))
            c3.metric("Company Trust", round(result["company_score"], 2))

            st.markdown("### 📉 Risk Contribution Breakdown")
            contrib_df = pd.DataFrame({"Component": ["Text Analysis", "Company Trust",
"Email Verification"],"Score": [result["text_score"],
                                                    1 -
result["company_score"],
                                                    #inverted →
```

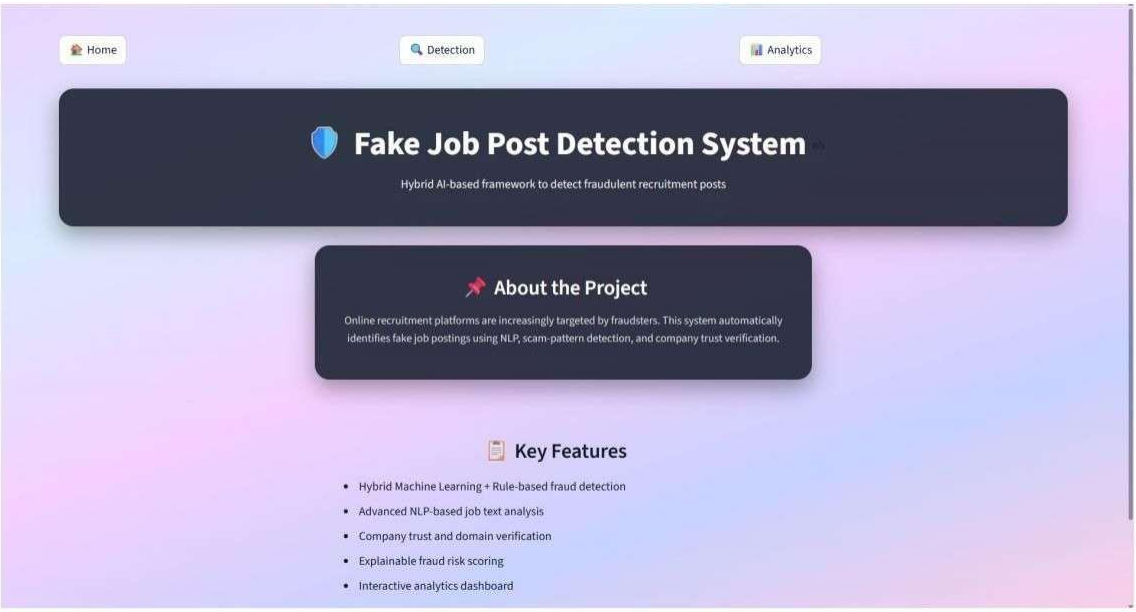
```
higher = risky

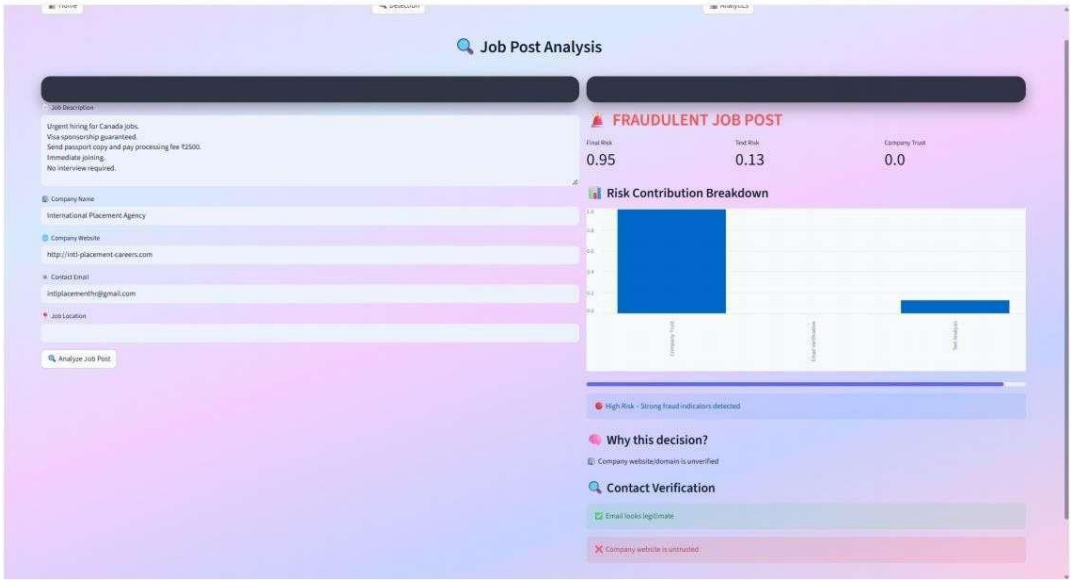
result["email_score"]]})
            st.bar_chart(contrib_df.set_index("Component"))
            st.progress(int(score * 100))
            st.info(interpret_risk(score))
            st.markdown("### ☐Why this decision?")
            reasons = []

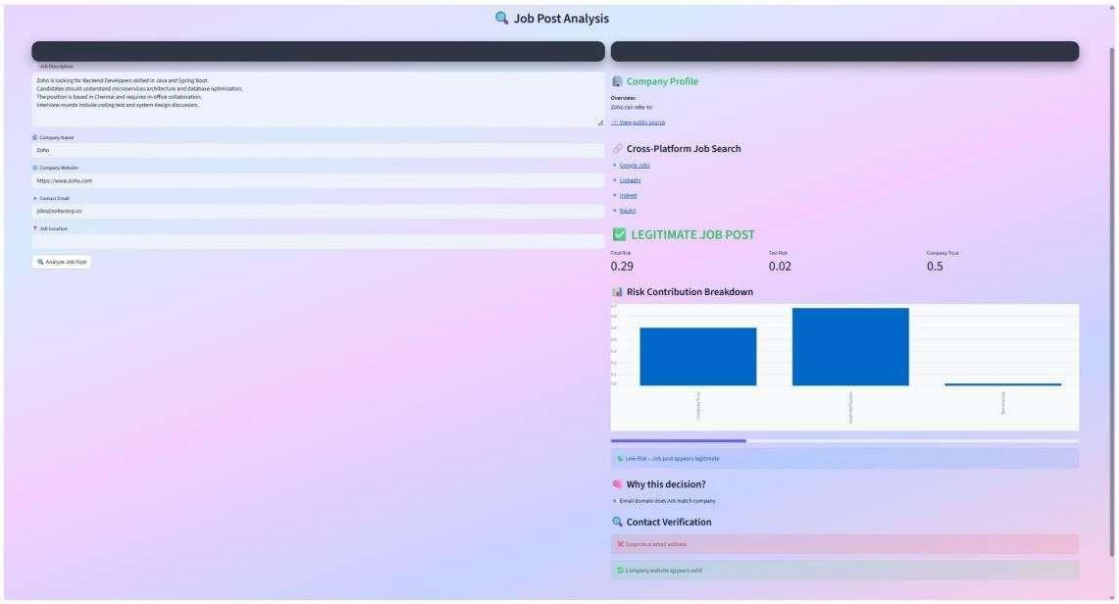
            if result["text_score"] > 0.6:
                reasons.append("📌 Scam-like phrases detected (urgency, high pay, no
interview)")
            if result["company_score"] < 0.4:
                reasons.append("🌐 Company website/domain is unverified")
            if result["email_score"] > 0.5:
                reasons.append("✉️ Email domain does not match company")
            if not reasons:
                reasons.append("✅ Job post follows standard hiring practices")

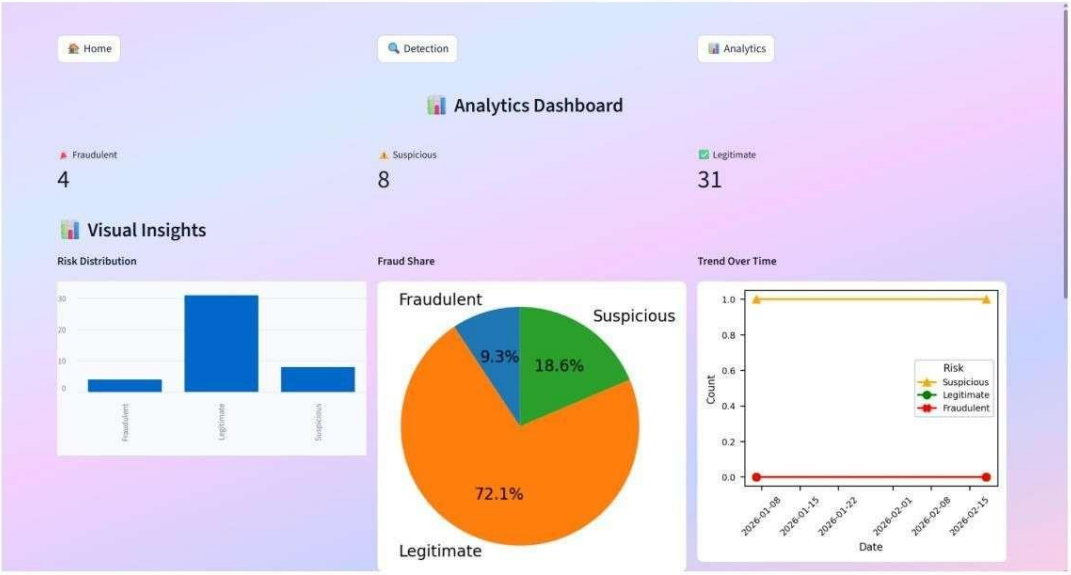
            for r in reasons:
                st.write(r)
            st.markdown("### 🔍 Contact Verification")
            if result["email_score"] < 0.5:
                st.success("✅ Email looks legitimate")
            else:
                st.error("❌ Suspicious email address")
            if result["company_score"] >= 0.4:
                st.success("✅ Company website appears valid")
            else:
                st.error("❌ Company website is untrusted")
            st.markdown("</div>", unsafe_allow_html=True)
    if stats:
        df = pd.DataFrame(stats, columns=["Risk Label", "Count"])
```

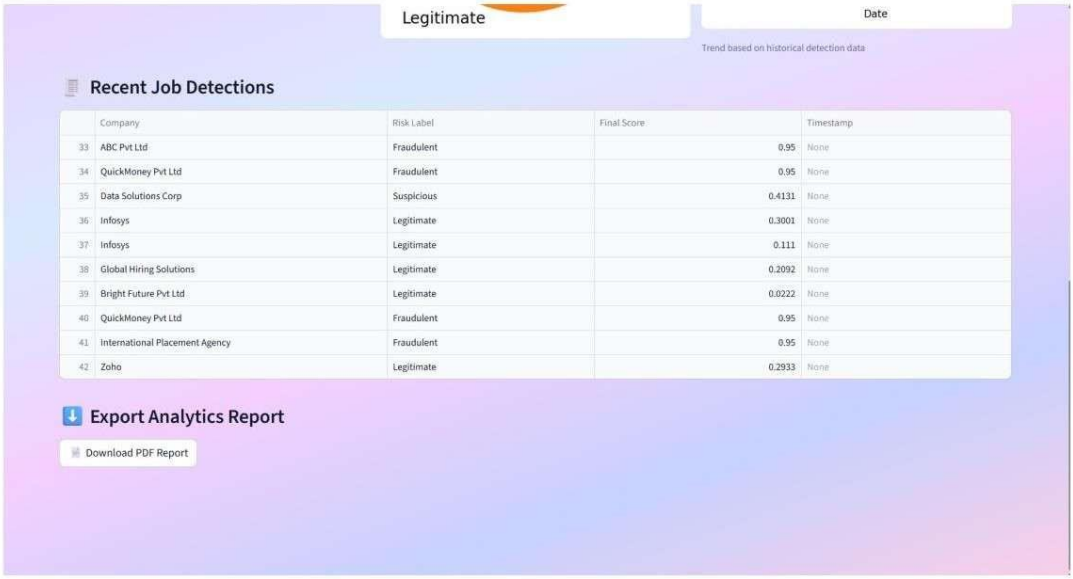
A.3 SAMPLE SCREENSHOTS






Job Post Analysis
Company Profile
Cross-Platform Job Search
LEGITIMATE JOB POST
0.29
0.02
0.5
Risk Contribution Breakdown
Why this decision?
Contact Verification


Home
Detection
Analytics
Analytics Dashboard
Fraudulent
4
Suspicious
8
Legitimate
31
Visual Insights
Risk Distribution
Fraud Share
Trend Over Time
Fraudulent
Suspicious
9.3%
18.6%
72.1%
Legitimate
Count
Date
Risk
Suspicious
Legitimate
Fraudulent


Legitimate
Date
Trend based on historical detection data
Recent Job Detections
Company
Risk Label
Final Score
Timestamp
33 ABC Pvt Ltd Fraudulent 0.95 None
34 QuickMoney Pvt Ltd Fraudulent 0.95 None
35 Data Solutions Corp Suspicious 0.4131 None
36 Infosys Legitimate 0.3001 None
37 Infosys Legitimate 0.111 None
38 Global Hiring Solutions Legitimate 0.2092 None
39 Bright Future Pvt Ltd Legitimate 0.0222 None
40 QuickMoney Pvt Ltd Fraudulent 0.95 None
41 International Placement Agency Fraudulent 0.95 None
42 Zoho Legitimate 0.2933 None
Export Analytics Report
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